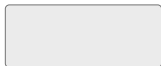


Confounding, Selection and Explainability

Weekend 2 · Introductory block

Carlos Cotrini

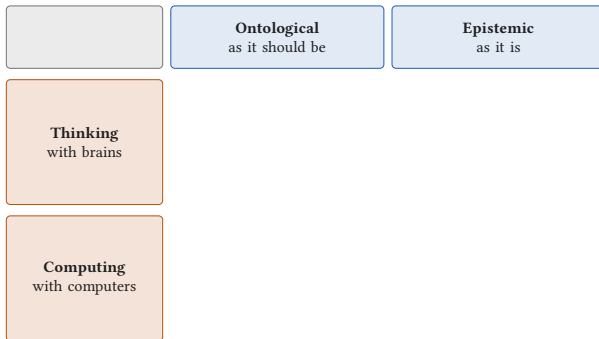
Four paradigms in science



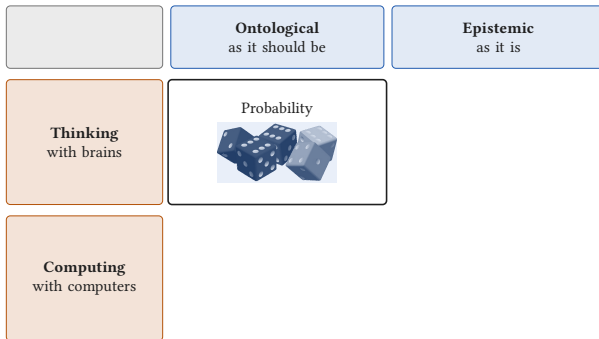
Ontological
as it should be

Epistemic
as it is

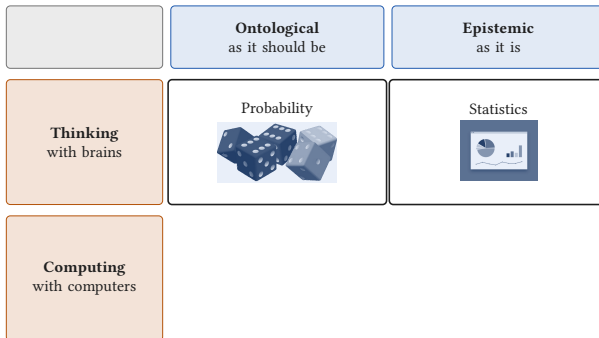
Four paradigms in science



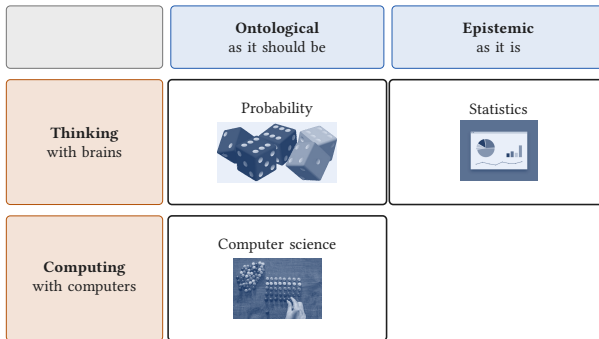
Four paradigms in science



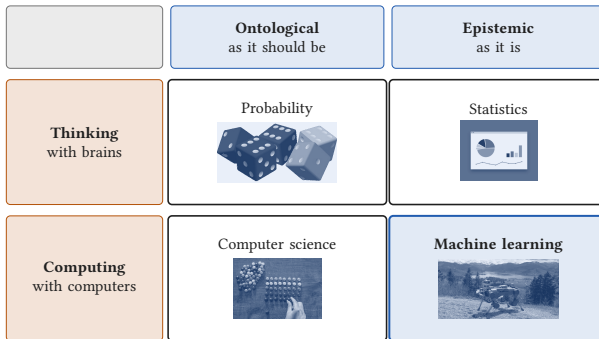
Four paradigms in science



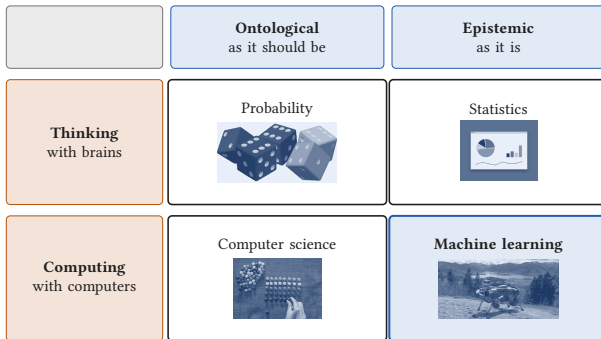
Four paradigms in science



Four paradigms in science



Four paradigms in science



It learns the world as it is: bad data misleads it.

Section 1

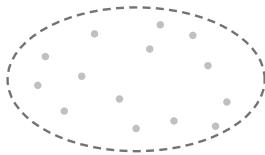
Statistical learning theory

1

Statistical learning theory

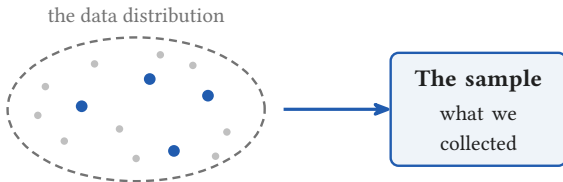
Proposed by Vladimir Vapnik.

the data distribution



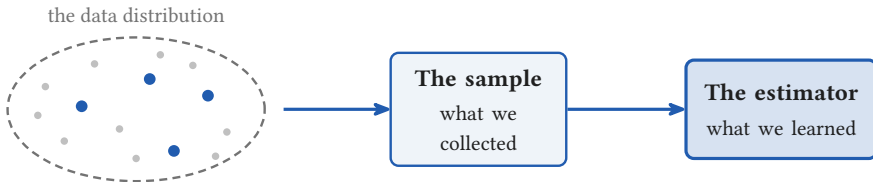
Statistical learning theory

Proposed by Vladimir Vapnik.



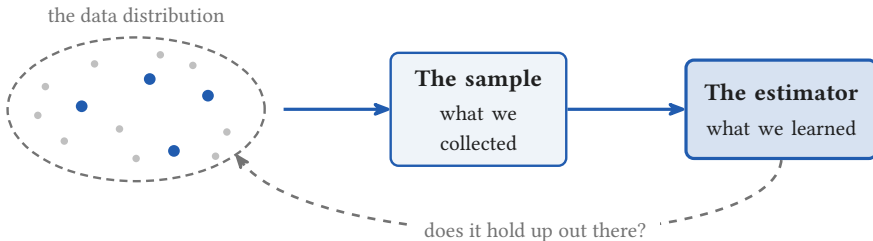
Statistical learning theory

Proposed by Vladimir Vapnik.



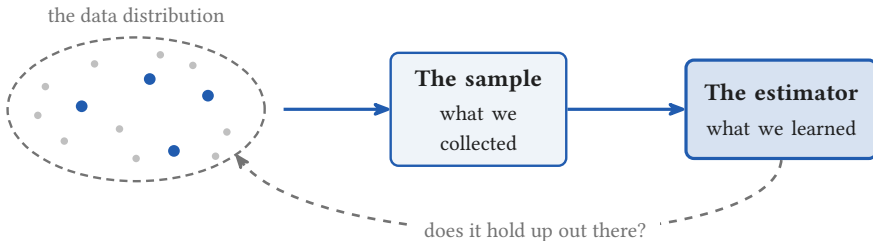
Statistical learning theory

Proposed by Vladimir Vapnik.



Statistical learning theory

Proposed by Vladimir Vapnik.



Learn from a sample, then ask whether it holds up

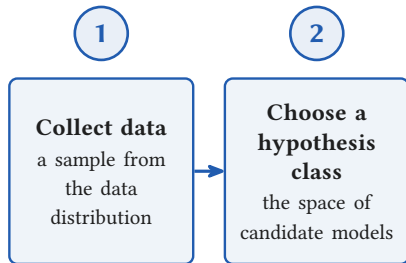
The five steps

1

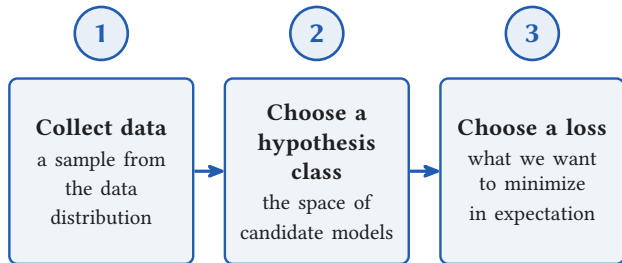
Collect data

a sample from
the data
distribution

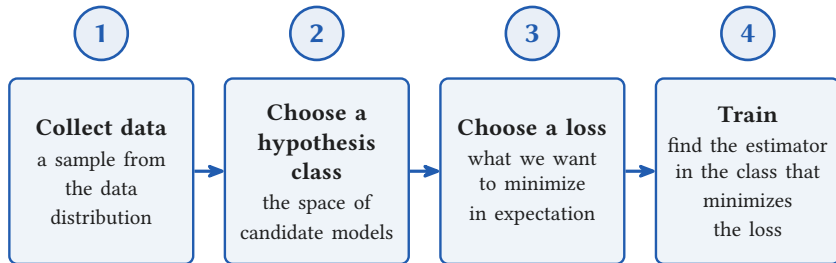
The five steps



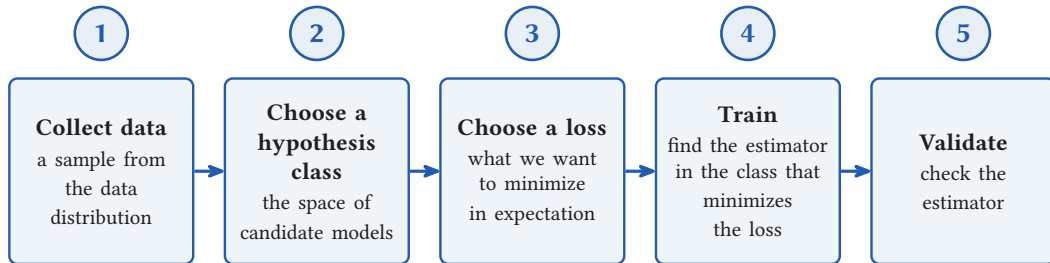
The five steps



The five steps



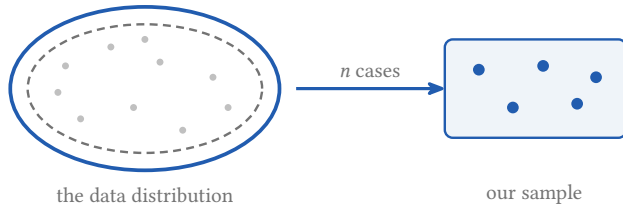
The five steps



Risk and empirical risk

$$R(f) = \mathbb{E} [\ell(f(X), Y)]$$

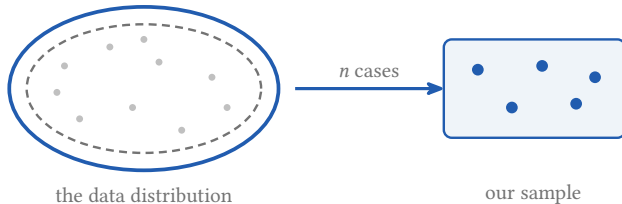
The risk: what we would like to minimize.



Risk and empirical risk

$$R(f) = \underbrace{\mathbb{E}_{(X,Y) \sim P}}_{\text{average over the distribution}} \left[\underbrace{\ell(f(X), Y)}_{\text{cost of one prediction}} \right]$$

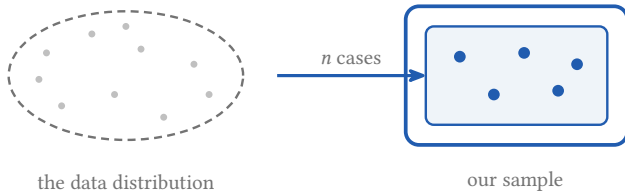
The risk: what we would like to minimize.



Risk and empirical risk

$$\hat{R}(f) = \frac{1}{n} \sum_{i=1}^n \ell(f(x_i), y_i)$$

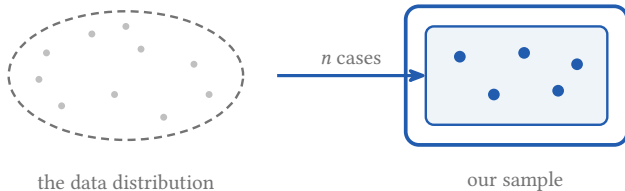
We only ever see a finite sample, so we minimize the empirical risk instead.



Risk and empirical risk

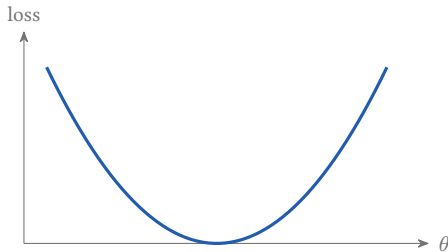
$$\widehat{R}(f) = \underbrace{\frac{1}{n} \sum_{i=1}^n}_{\text{average over our sample}} \underbrace{\ell(f(x_i), y_i)}_{\text{cost on case } i}$$

We only ever see a finite sample, so we minimize the empirical risk instead.



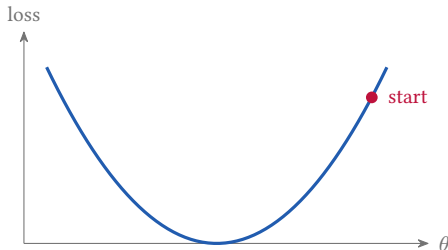
Training in practice: gradient descent

- Step 4 (train) is usually **gradient descent**.



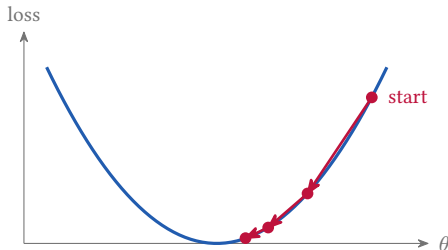
Training in practice: gradient descent

- Step 4 (train) is usually **gradient descent**.
- The loss is a landscape. We walk downhill.



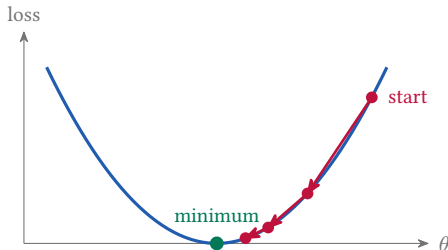
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- Each step moves against the gradient.



Training in practice: gradient descent

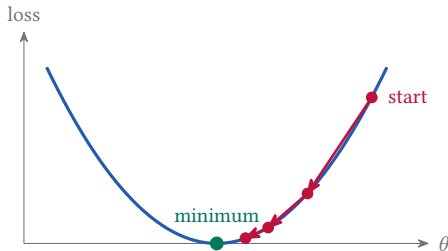
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Training in practice: gradient descent

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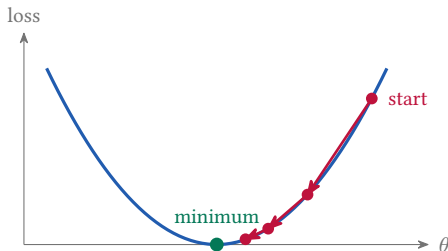
$$\theta \leftarrow \theta - \eta \nabla L(\theta)$$



Training in practice: gradient descent

- Step 4 (train) is usually **gradient descent**.
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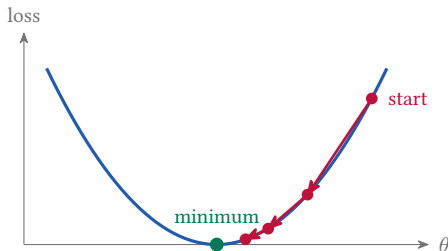
$$\theta \leftarrow \theta - \underbrace{\eta}_{\text{step size}} \underbrace{\nabla L(\theta)}_{\text{uphill direction}}$$



Training in practice: gradient descent

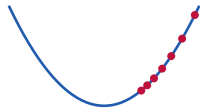
- Step 4 (train) is usually **gradient descent**.
- The loss is a landscape. We walk downhill.
- Each step moves against the gradient.

$$\theta \leftarrow \theta - \underbrace{\eta}_{\text{step size}} \underbrace{\nabla L(\theta)}_{\text{uphill direction}}$$



Follow the slope down to the smallest loss

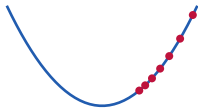
The one knob: the learning rate



Too small

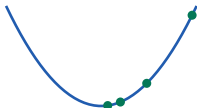
it crawls

The one knob: the learning rate



Too small

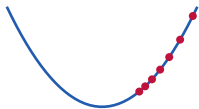
it crawls



Just right

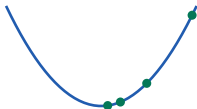
it converges

The one knob: the learning rate



Too small

it crawls



Just right

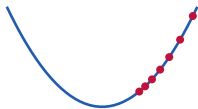
it converges



Too big

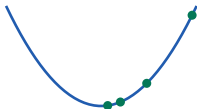
it overshoots

The one knob: the learning rate



Too small

it crawls



Just right

it converges

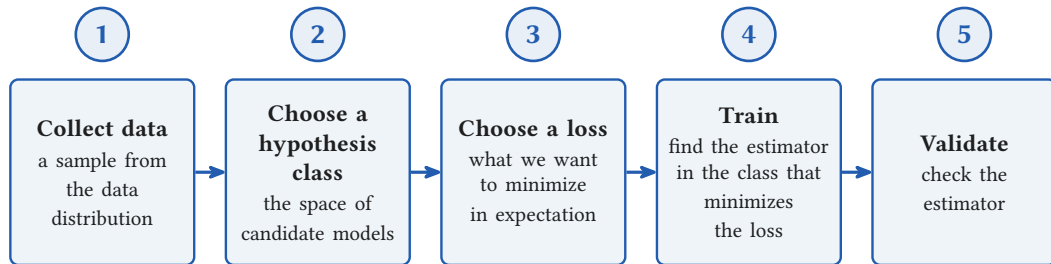


Too big

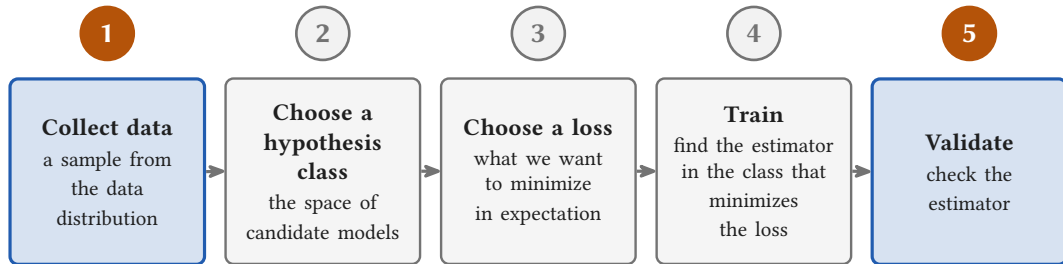
it overshoots

The step size makes or breaks the descent

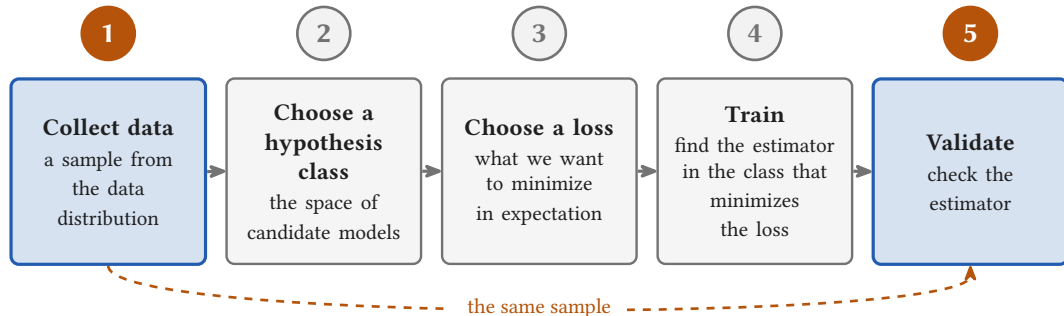
The two steps this weekend attacks



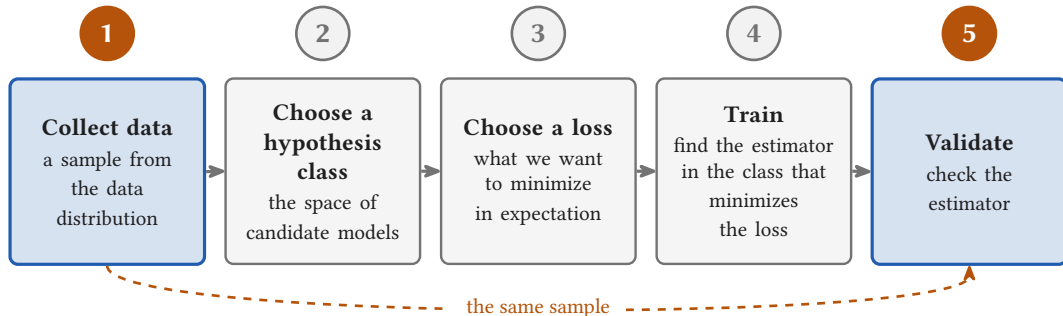
The two steps this weekend attacks



The two steps this weekend attacks



The two steps this weekend attacks



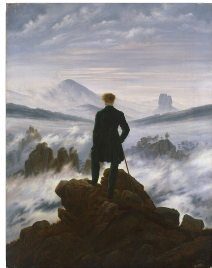
If the sample lies, validation will happily agree

Section 2

The Island of Truth

2

A land of truth in a sea of illusion

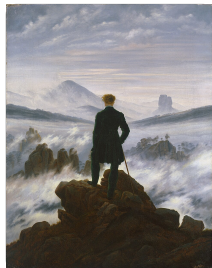


C. D. Friedrich,

Wanderer above the Sea of Fog (c. 1817)

Es ist das *Land der Wahrheit* (ein reizender Name), umgeben von einem weiten und stürmischen Oceane, dem eigentlichen Sitze des Scheins, wo manche Nebelbank und manches bald wegschmelzende Eis neue Länder lügt ...

A land of truth in a sea of illusion



C. D. Friedrich,

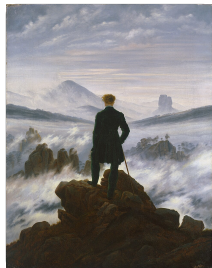
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It is the *land of truth* (an enchanting name), surrounded by a wide and stormy ocean, the very seat of illusion, where many a fog bank and many a swiftly melting iceberg feign new lands ...

Kritik der reinen Vernunft, A235 / B294 (1781 / 1787)

A land of truth in a sea of illusion



C. D. Friedrich,

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Weekend 1 was the fog. This weekend is the ice.

The fog and the ice



Weekend 1

Fog

Honest uncertainty.

We know that we cannot see.

$H[X]$, $\text{Var}[X]$

The fog and the ice

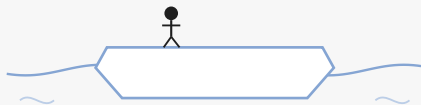


Weekend 1

Fog

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This weekend

Ice

It looks like solid ground.
It holds on training and on test.

The fog and the ice

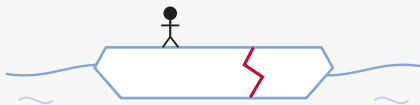


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The fog and the ice

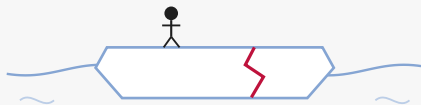


Weekend 1

Fog

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This weekend

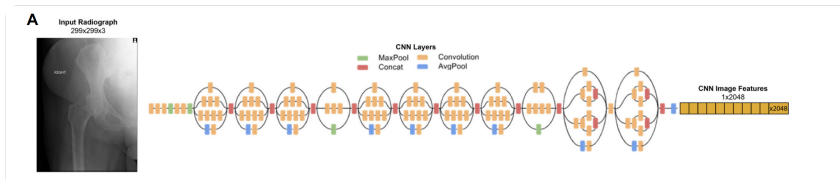
Ice

It looks like solid ground.
It holds on training and on test.

Then it breaks.

The ice: a false conclusion, or an estimator that fails to generalize.

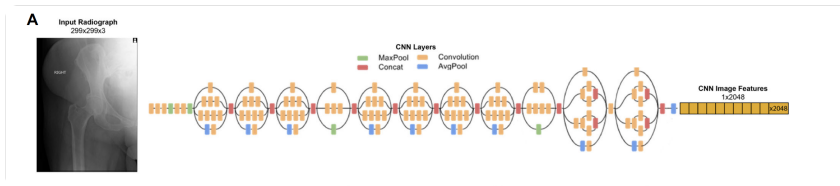
Predicting hip fracture from X-rays



- 23,602 radiographs, Mount Sinai, 2008 to 2016 (3% with a fracture)

Architecture from Badgeley et al., [npj Digital Medicine 2:31 \(2019\)](#).

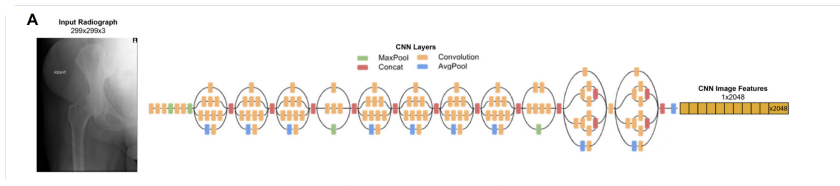
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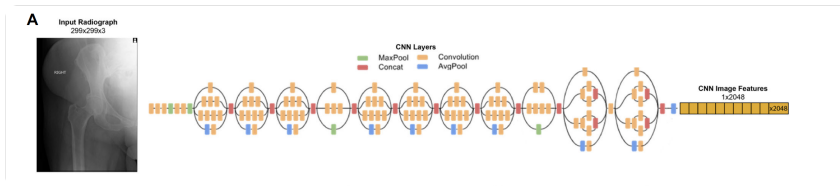
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Predicting hip fracture from X-rays

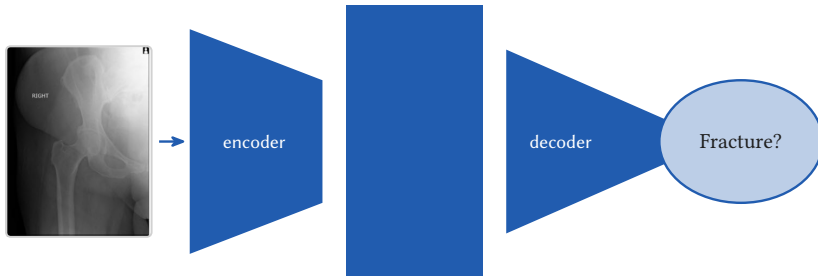


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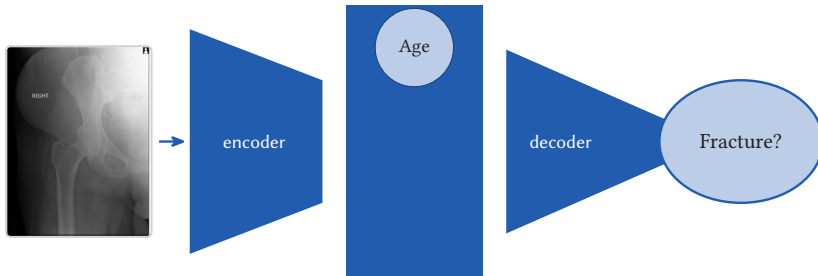
Sounds good.

Architecture from Badgeley et al., [npj Digital Medicine](#) 2:31 (2019).

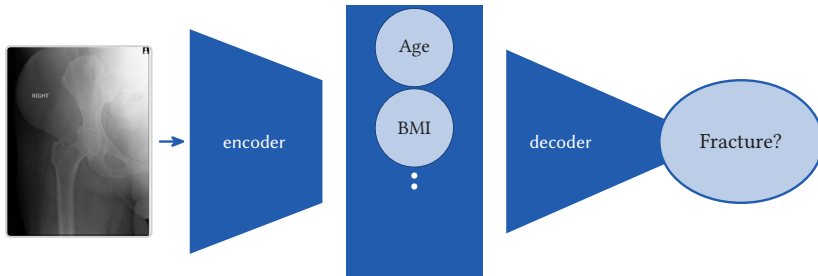
What did the network actually learn?



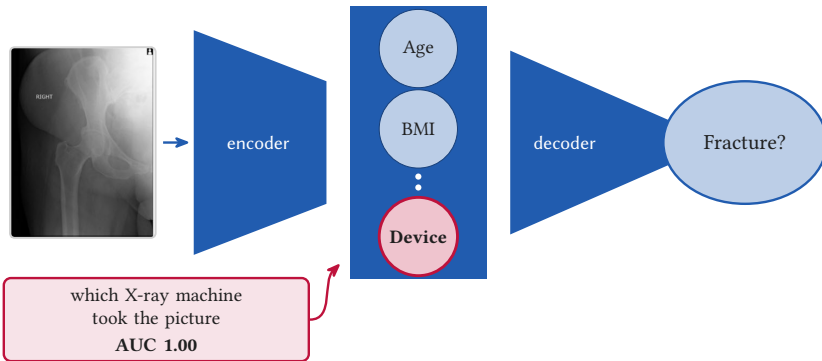
What did the network actually learn?



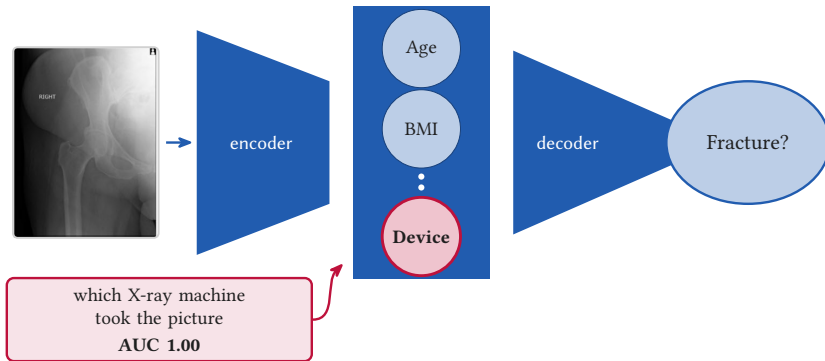
What did the network actually learn?



What did the network actually learn?

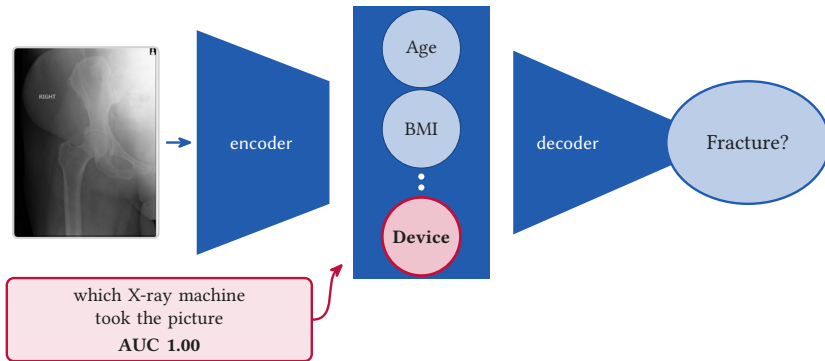


What did the network actually learn?



The model read the scanner, not the bone.

What did the network actually learn?



The model read the scanner, not the bone.

Different machines served different patient populations, so the scanner tracks the fracture rate.

The image features cluster by scanner

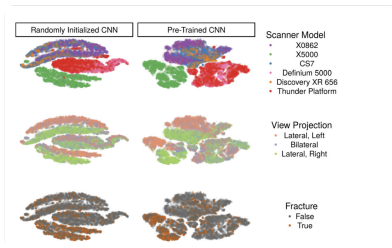


Image features, colored by scanner (top),
view (middle), fracture (bottom).



[Badgeley et al. \(2019\)](#). The evidence is statistical, not pixel level: the paper reports no saliency map, and the collapse comes from matching the confounders away.

The image features cluster by scanner

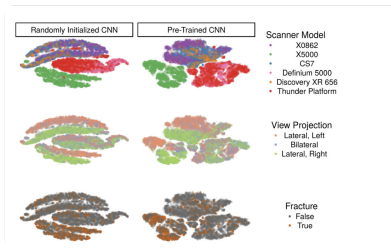
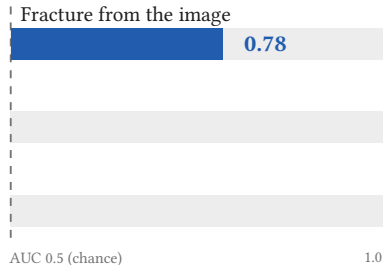


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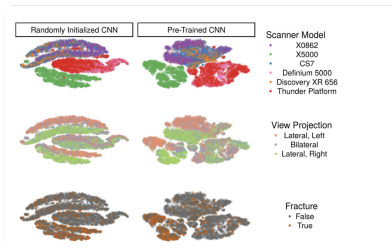
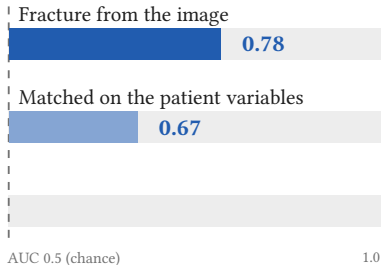


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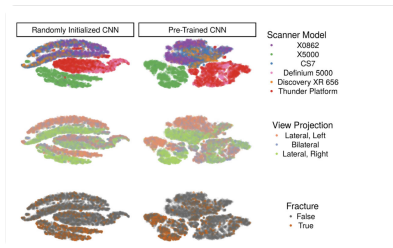
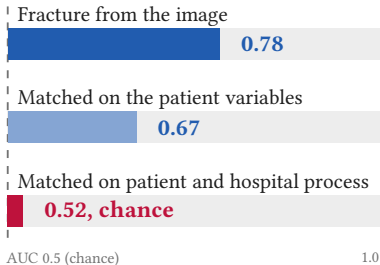


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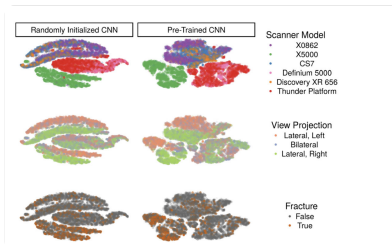
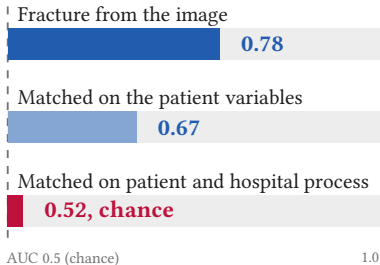


Image features, colored by scanner (top),
view (middle), fracture (bottom).

Hospital process variables alone, with no image at all: AUC 0.89.



Badgeley et al. (2019). The evidence is statistical, not pixel level: the paper reports no saliency map, and the collapse comes from matching the confounders away.

It worked on the training set.

It worked on the test set.

It fails in a new hospital.

It is not one bad paper

Chest X-rays: pneumonia

Reads the hospital off the film: 99.9%. One cue: a metal token in the corner. Pneumonia: 34.2% at one site, 1.2% at the other.

[Zech et al., PLOS Medicine \(2018\)](#)

It is not one bad paper

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Mammograms: breast cancer

Six models, reported auROC up to 0.95. On three external data sets: 0.44 to 0.65.

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Wang et al., J. Am. Coll. Radiol. (2020)

Chest X-rays: COVID

Internal AUC 0.99, external 0.76. Swap the laterality marker between two films: the diagnosis moves.

DeGrave et al., Nature Mach. Intell. (2021)

It is not one bad paper

Chest X-rays: pneumonia

Reads the hospital off the film: 99.9%. One cue: a metal token in the corner. Pneumonia: 34.2% at one site, 1.2% at the other.

Zech et al., PLOS Medicine (2018)

Mammograms: breast cancer

Six models, reported auROC up to 0.95. On three external data sets: 0.44 to 0.65.

Wang et al., J. Am. Coll. Radiol. (2020)

Chest X-rays: COVID

Internal AUC 0.99, external 0.76. Swap the laterality marker between two films: the diagnosis moves.

DeGrave et al., Nature Mach. Intell. (2021)

The whole literature

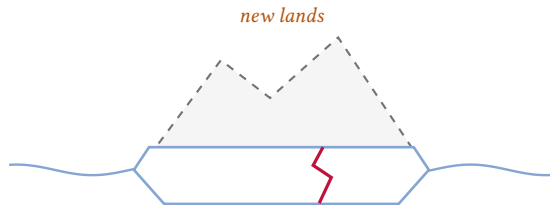
2,212 COVID imaging studies screened, 62 reviewed. None of the models is fit for clinical use.

Roberts et al., Nature Mach. Intell. (2021)

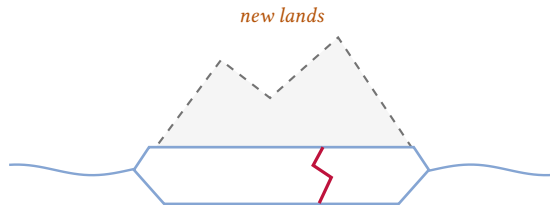
Where this leaves us



Where this leaves us



Where this leaves us



The ice is a model that generalizes to your test set, and to nothing else.

This weekend

- 1** **Recap**
where Weekend 1 left us
- 2** **Confounding and selection**
the two ways data misleads a model
- 3** **Explainable AI: attribution maps**
tools to see what a model used
- 4** **Prompt optimization (GEPA)**
a separate module

Section 3

How humans step on the ice

3

Reverse Causality



“Windmills create wind!”

Reverse Causality



“Windmills create wind!”

They are built where the wind already blows.

Reverse Causality



“Windmills create wind!”

They are built where the wind already blows.



“Lice are healthy for you!”

Reverse Causality



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“Lice are healthy for you!”

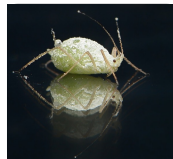
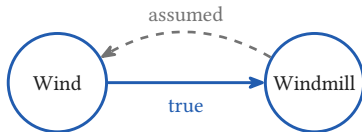
Lice leave a body that runs a fever.

Reverse Causality



“Windmills create wind!”

They are built where the wind already blows.



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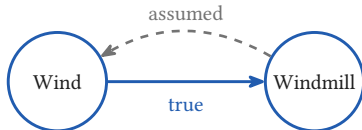
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“Windmills create wind!”

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“Lice are healthy for you!”

Lice leave a body that runs a fever.

$$\text{Corr}(X, Y) = \text{Corr}(Y, X)$$

The correlation is symmetric. The arrow is not.

The Third-Cause Fallacy

Young children who sleep with the light on are much more likely to develop myopia in later life.



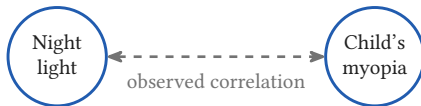
Published in *Nature* in 1999.

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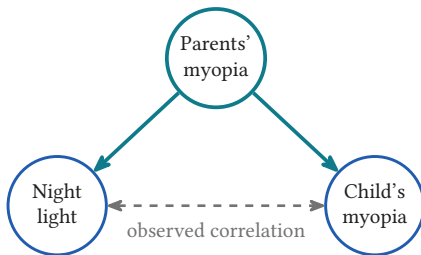


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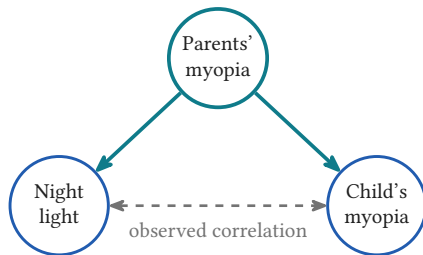


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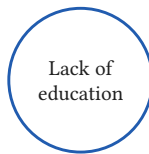
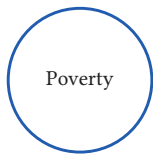
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Myopic parents leave the light on, and pass on the myopia.

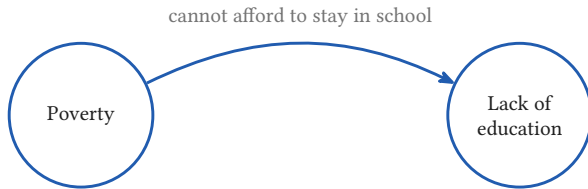
Bidirectional Causation

Poverty and lack of education.



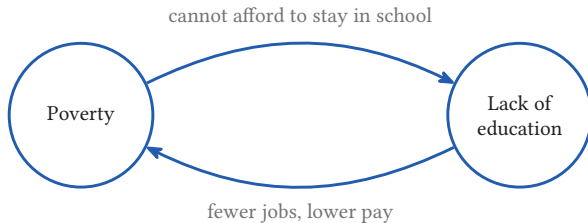
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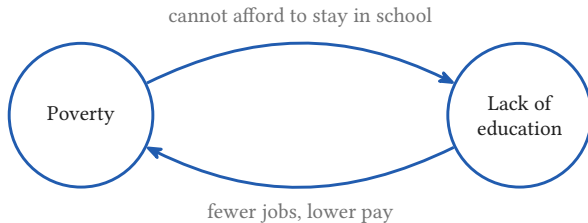
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Bidirectional Causation

Poverty and lack of education.



Both directions are real. One coefficient cannot separate them.

Spurious Relation

Mozzarella cheese consumption and civil engineering doctorates awarded.

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Mozzarella cheese consumption and civil engineering doctorates awarded.

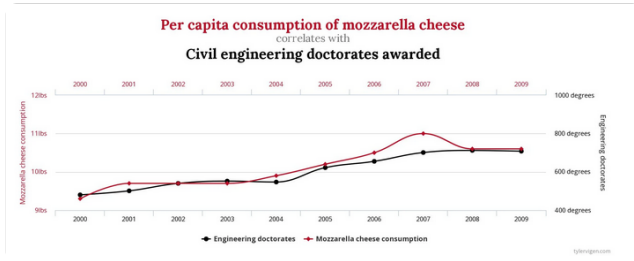


Chart: tylervigen.com, Spurious Correlations.

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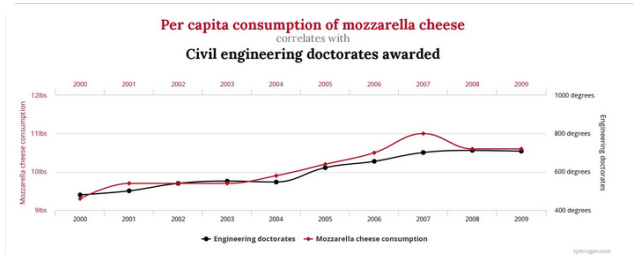


Chart: tylervigen.com, Spurious Correlations.

With enough series, something will always line up.

Spurious Correlations Are Everywhere

Real data. Real correlations. No causal link whatsoever.

- Films with **Nicolas Cage** vs. **drownings in a pool**

$$r = 0.67$$

tylervigen.com, roughly 1999 to 2009.

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Screen millions of pairs, and something always lines up

The Four Fallacies

Reverse causality

the arrow points back



The Four Fallacies

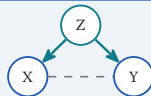
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Third cause

a hidden Z causes both



The Four Fallacies

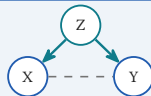
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Bidirectional causation

each causes the other



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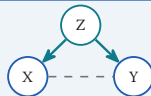
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no link, a coincidence



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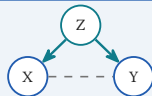
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Bidirectional causation

each causes the other



Spurious relation

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Every correlation was real.
Only the causal story was wrong.

Are These Four All?

The four blamed a wrong causal *story*. But often the correlation itself is an [artifact](#), from how the data was sampled, aggregated, or searched.

The causal story is wrong

reverse, third cause,
bidirectional, coincidence

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reverse, third cause,
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The four were only half the story

More Fallacies

- **Regression to the mean:** extremes drift back on their own.

Praise a great landing and the next is worse; scold a bad one and the next is better. Chance, not you.

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And three we spend the rest of today on: confounding, selection, and Simpson's paradox.

Section 4

Machines step on the ice too

4

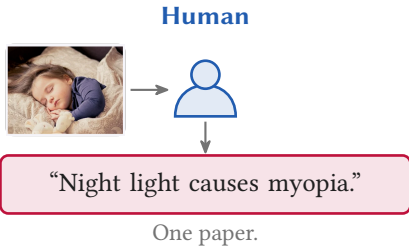
A model makes the same mistake, faster



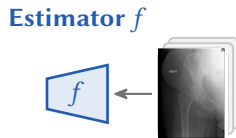
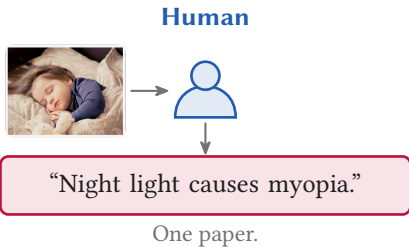
Human



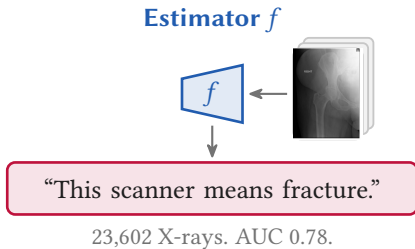
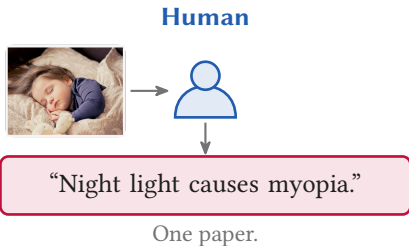
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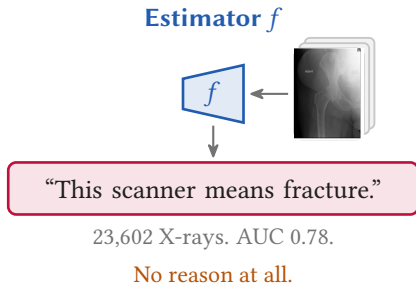
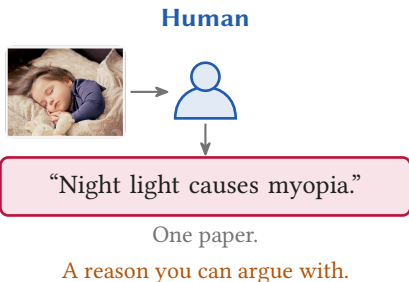
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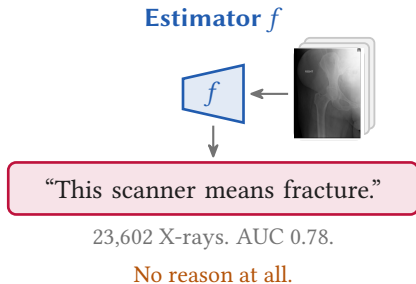
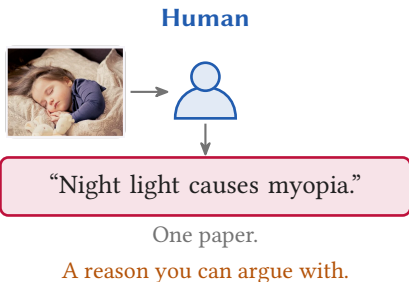


A model makes the same mistake, faster



Hip fracture: Badgeley et al., npj Digital Medicine, 2019. Night light: Nature, 1999.

A model makes the same mistake, faster



Hip fracture: Badgeley et al., npj Digital Medicine, 2019. Night light: Nature, 1999.

Same fallacy. At scale. And silent.

**We want the model
off the ice.**

**First we need a language
for why it steps on it:
causal graphs.**

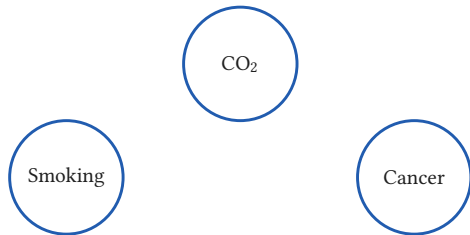
Section 5

Causal graphs

5

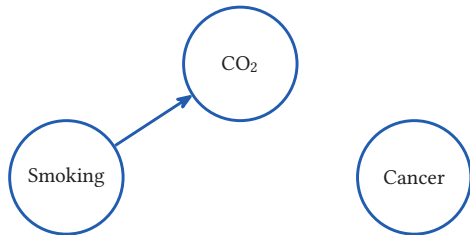
Causal graphs

- Nodes: the variables.



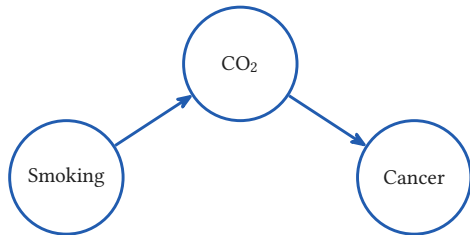
Causal graphs

- Nodes: the variables.
- An arrow: a direct cause.



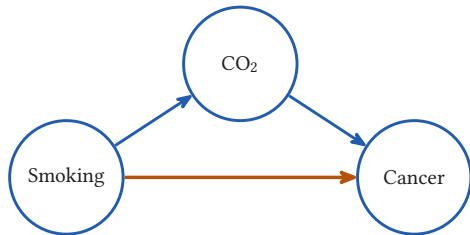
Causal graphs

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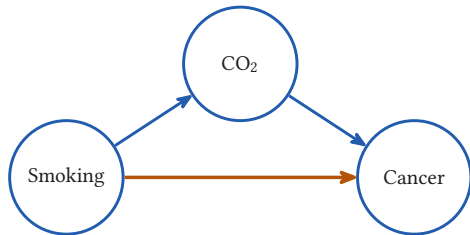
Causal graphs

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- A path: an indirect cause.
- A variable can act both ways.



Causal graphs

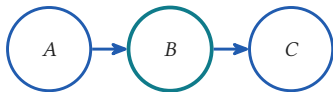
- Nodes: the variables.
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Arrows flow one way: no cycles.

Reading a graph

Chain

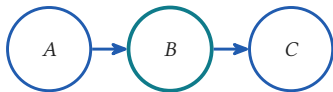


B is a **mediator**

A acts through *B*

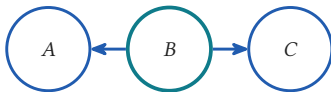
Reading a graph

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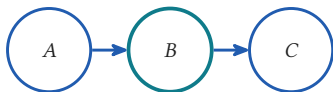
Fork



B is a **common cause**
a confounder

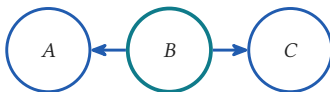
Reading a graph

Chain



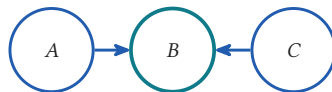
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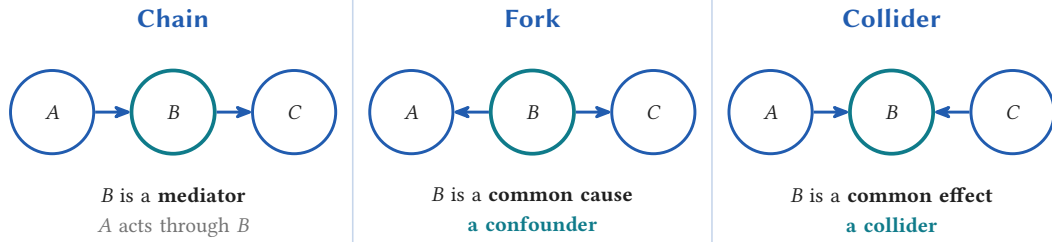
*B is a **common cause**
a **confounder***

Collider



*B is a **common effect**
a **collider***

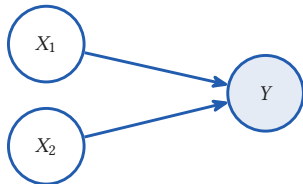
Reading a graph



Same three nodes. Only the arrows change.

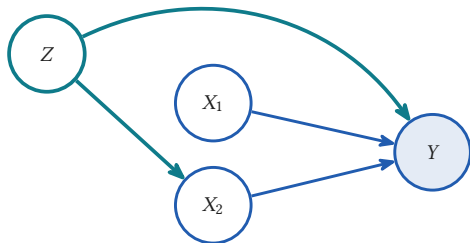
Why we draw the graph

- How features reach the target.



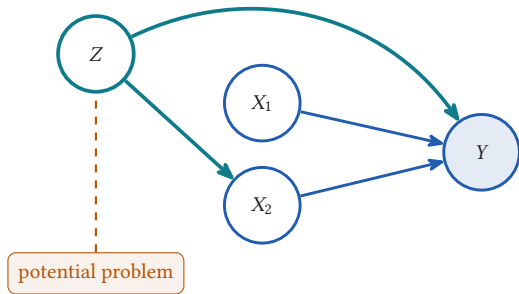
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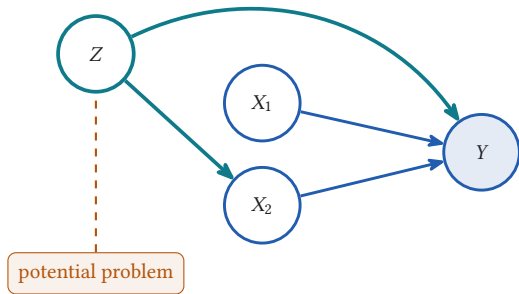
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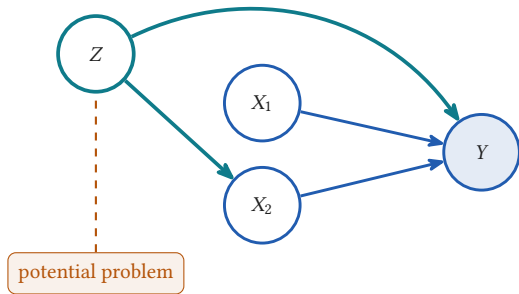
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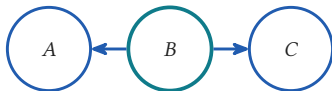
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- Not which problems are real.



An assumption, not a proof.

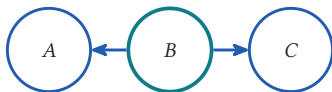
Two traps, in opposite directions

Of the three shapes, two are traps.



Two traps, in opposite directions

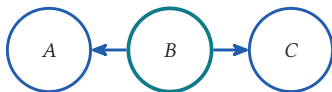
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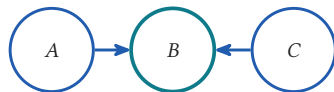
Fork
control for *B*

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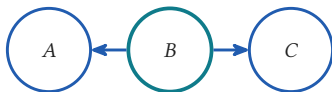


Fork
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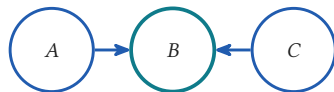


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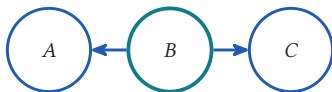
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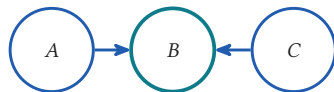
Collider
do **not** control for *B*

Two traps, in opposite directions

Of the three shapes, two are traps.



Fork
control for *B*



Collider
do **not** control for *B*

Two traps. Opposite fixes.

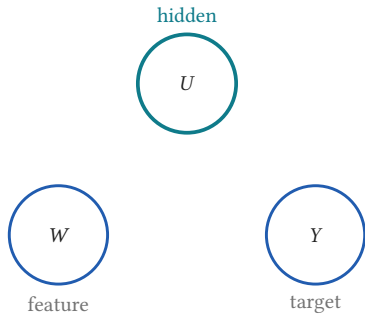
Next: what each one does to us.

Section 6

Confounding

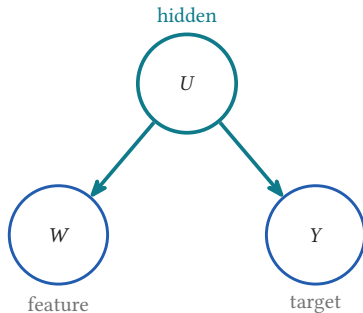
6

Confounding



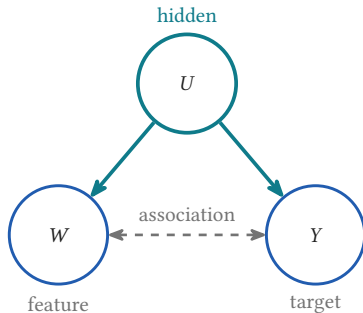
Confounding

- Hidden U causes both W and Y .



Confounding

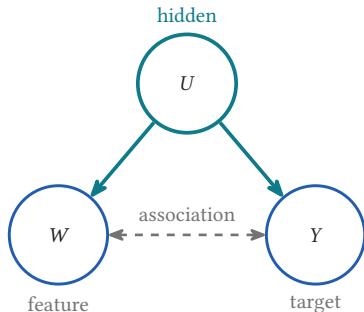
- Hidden U causes both W and Y .
- No arrow from W to Y .



Confounding

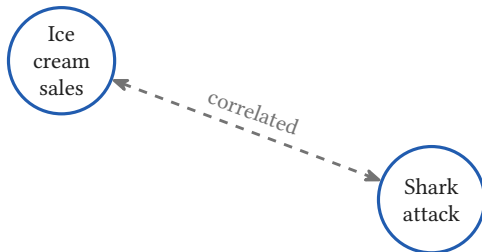
- Hidden U causes both W and Y .
- No arrow from W to Y .
- The estimator fits the association anyway.

$$W \leftarrow U \rightarrow Y \implies P(Y | W) \neq P(Y)$$



Confounding: Ice Cream and Shark Attacks

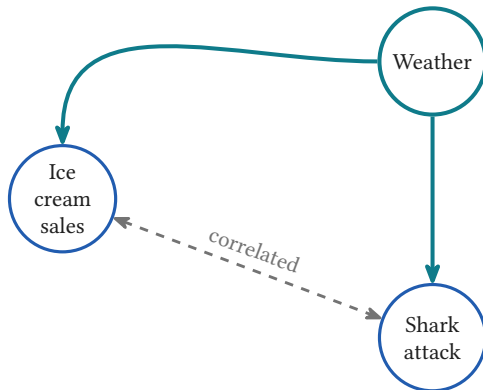
Observed
sales predict attacks.



Confounding: Ice Cream and Shark Attacks

Observed
sales predict attacks.

Hidden common cause
the weather.

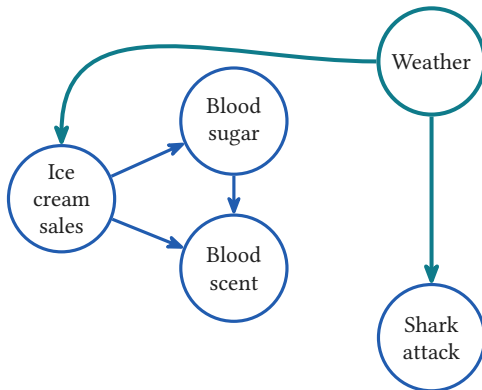


Confounding: Ice Cream and Shark Attacks

Observed
sales predict attacks.

Hidden common cause
the weather.

Blue edges
the mechanism the graph
draws.

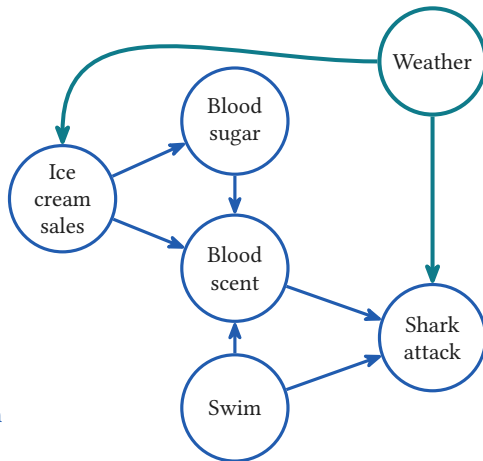


Confounding: Ice Cream and Shark Attacks

Observed
sales predict attacks.

Hidden common cause
the weather.

Blue edges
the mechanism the graph
draws.



Simpson's Paradox

	Treatment A open surgery	Treatment B PCNL

Charig et al., BMJ 1986. The size stratified table is [Julious and Mullee, BMJ 1994](#).

Simpson's Paradox

	Treatment A open surgery	Treatment B PCNL
Overall	$\frac{273}{350} = 78\%$	$\frac{289}{350} = 83\%$

Overall, Treatment B looks better.

Charig et al., BMJ 1986. The size stratified table is Julious and Mullee, BMJ 1994.

Simpson's Paradox

	Treatment A open surgery	Treatment B PCNL
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Small stones	$\frac{81}{87} = 93\%$	$\frac{234}{270} = 87\%$

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But by stone size, A wins every time.

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Simpson's Paradox

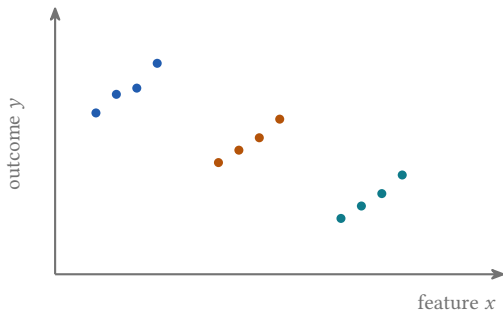
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Large stones	$\frac{192}{263} = 73\%$	$\frac{55}{80} = 69\%$

But by stone size, A wins every time.

The severe cases went to A: 263 of its 350 patients had large stones, against 80 of B's.

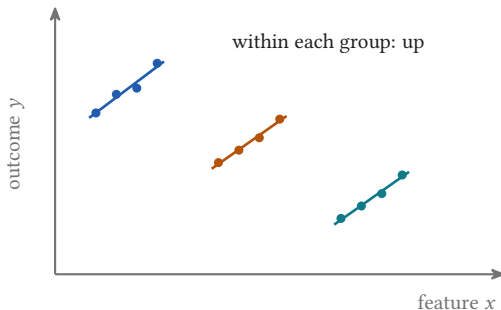
Charig et al., BMJ 1986. The size stratified table is Julious and Mullee, BMJ 1994.

Simpson's Paradox: An Interactive Explorer



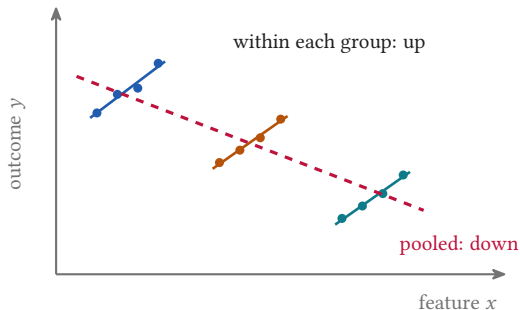
The explorer lets you drag the groups and watch the pooled line flip sign.

Simpson's Paradox: An Interactive Explorer



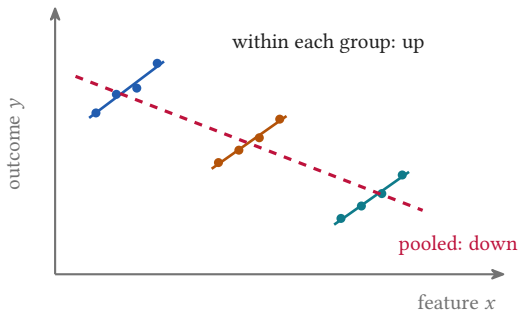
The explorer lets you drag the groups and watch the pooled line flip sign.

Simpson's Paradox: An Interactive Explorer



The explorer lets you drag the groups and watch the pooled line flip sign.

Simpson's Paradox: An Interactive Explorer

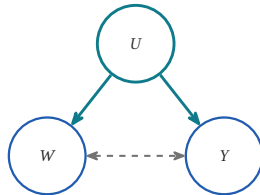


The explorer lets you drag the groups and watch the pooled line flip sign.

Try it on the course site.

Confounding in the Wild

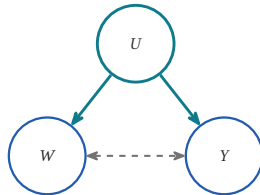
- **Hormone therapy** (healthy users): cohorts saw RR 0.56, the randomized trial found CHD hazard ratio 1.29. [WHI](#), [Rossouw et al.](#), [JAMA 2002](#)



The same fork, four times.

Confounding in the Wild

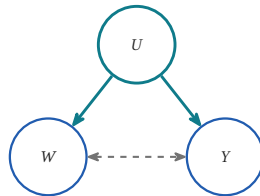
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- **Berkeley 1973 (department):** 44% of men admitted against 35% of women, reversed department by department. [Bickel et al.](#), [Science 1975](#)



The same fork, four times.

Confounding in the Wild

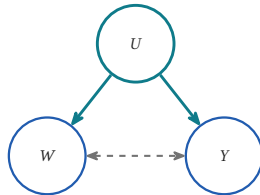
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- **Care algorithm (cost, not need):** fixing the label raised Black patients flagged from 17.7% to 46.5%. [Obermeyer et al.](#), [Science 2019](#)



The same fork, four times.

Confounding in the Wild

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- **Care algorithm (cost, not need):** fixing the label raised Black patients flagged from 17.7% to 46.5%. [Obermeyer et al.](#), [Science 2019](#)
- **Smoking (a gene?):** Fisher's counter to Doll and Hill; Bradford Hill answered with his viewpoints. [Fisher](#), [Nature 1958](#)



The same fork, four times.

Section 7

Selection

7

Selection

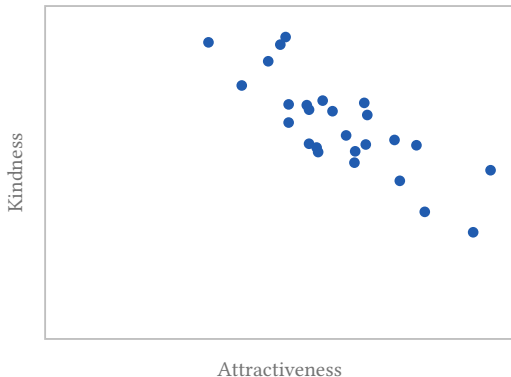
- Mary scores every past date on attractiveness and kindness.



Simulated data.

Selection

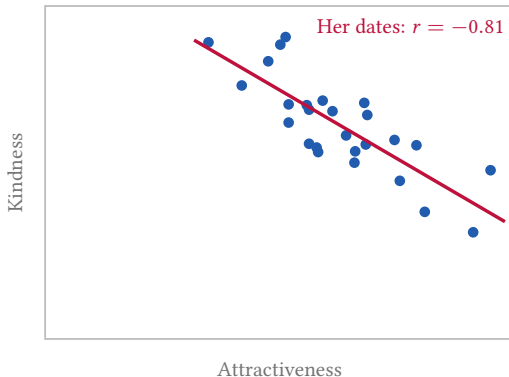
- Mary scores every past date on **attractiveness** and **kindness**.
- Her correlation analysis: the more attractive dates were **less** **kind**.



Simulated data.

Selection

- Mary scores every past date on **attractiveness** and **kindness**.
- Her correlation analysis: the more attractive dates were **less kind**.
- She decides she is kind enough, and stops worrying about her looks.

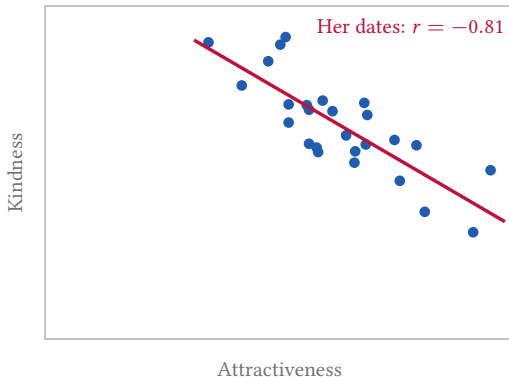


Simulated data.

Selection

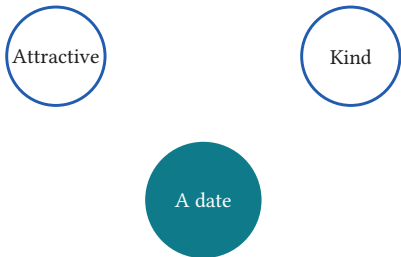
- Mary scores every past date on **attractiveness** and **kindness**.
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- She decides she is kind enough, and stops worrying about her looks.

Should she believe it?



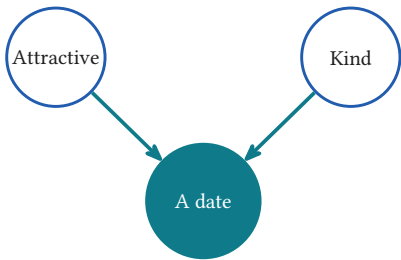
Simulated data.

Selection: A Collider



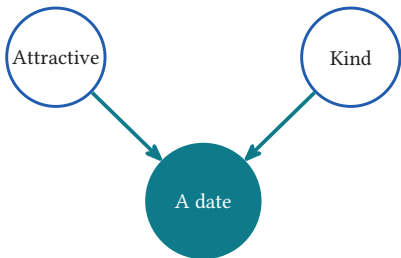
- Mary only ever sees the people she went on a date with.

Selection: A Collider



- Mary only ever sees the people she went on a date with.
- You get a date by being attractive, *or* by being kind.

Selection: A Collider

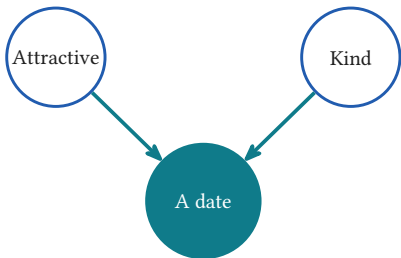


No arrow between attractiveness and kindness.

Filled: the node we conditioned on.

- Mary only ever sees the people she went on a date with.
- You get a date by being attractive, *or* by being kind.
- So a plain-looking date is probably there for the other reason.

Selection: A Collider



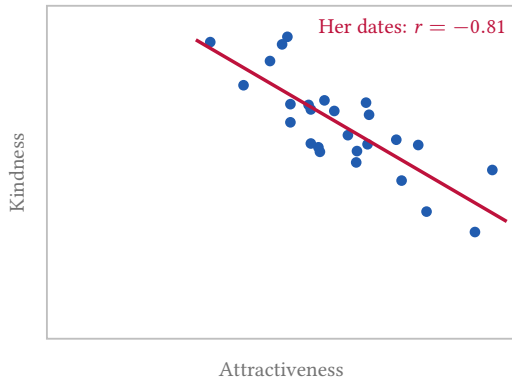
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- So a plain-looking date is probably there for the other reason.

The sampling manufactured the correlation.

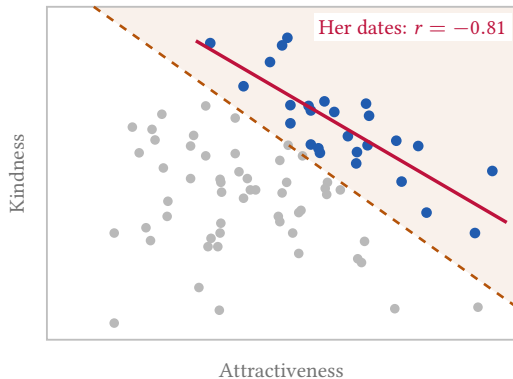
The Population Mary Never Saw



- The 26 people she dated (blue).

Simulated data. A teaching device (Ellenberg, 2014), not a measured result.

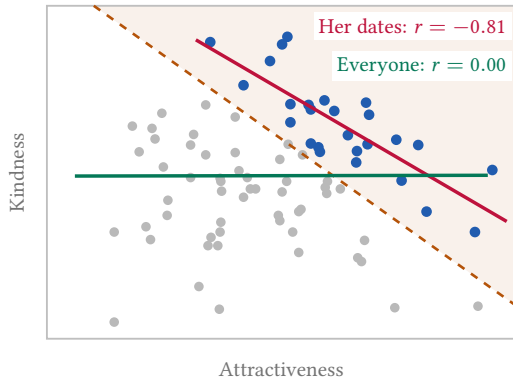
The Population Mary Never Saw



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- The 54 she never did, and **the bar**: attractive *or* kind gets you a date.

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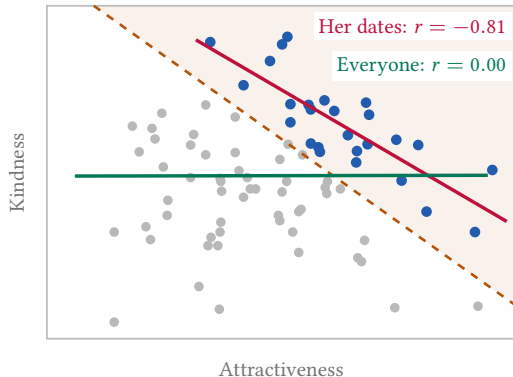
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The Population Mary Never Saw



- The 26 people she dated (blue).
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- In the whole population, no relation at all.

Drag the bar. Watch the slope.

[the collider explorer](#)

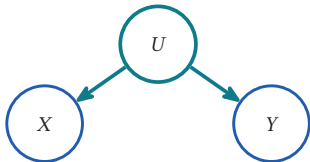
Simulated data. A teaching device (Ellenberg, 2014), not a measured result.

Selection Versus Confounding

Both U and S are correlated with X and Y .

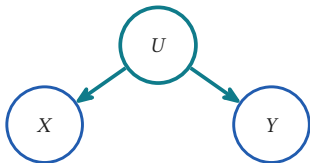
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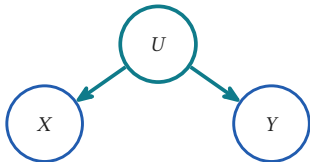
Weather drives ice cream sales *and* shark attacks.

Fork: control for U .

Leave it out and the link is fake.

Selection Versus Confounding

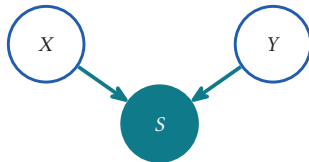
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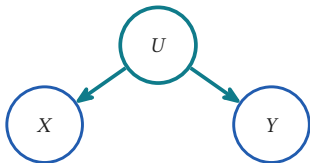
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Leave it out and the link is fake.



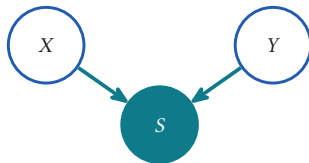
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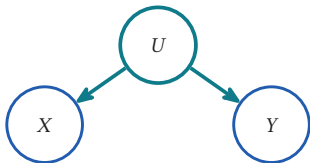


Attractiveness *and* kindness both get you a date with Mary.

Collider: do not control for S .
Put it in and the link is fake.

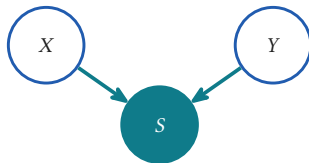
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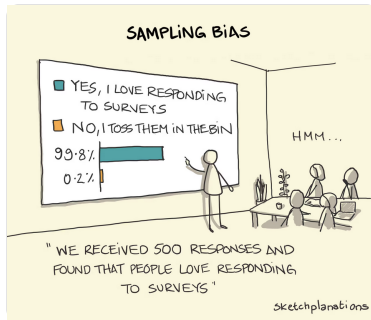
Collider: do not control for S .
Put it in and the link is fake.

Only the causal graph says which is which.

Selection in the Wild: Missing Rows

- **Wald, 1943.** The bombers that came home were hit everywhere but the engines. So armour the engines.

[Mangel and Samaniego, JASA 1984](#). The dotted plane came later.

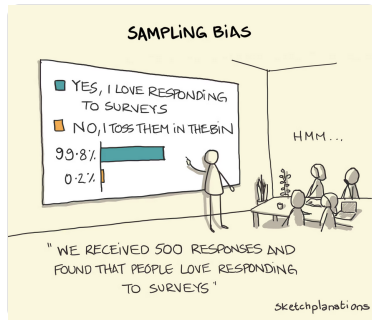


sketchplanations.com/sampling-bias

Selection in the Wild: Missing Rows

- **Wald, 1943.** The bombers that came home were hit everywhere but the engines. So armour the engines.
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- **Challenger, 1986.** They reviewed only the flights that already had O-ring damage. The cold weather signal vanished.

[Dalal, Fowlkes and Hoadley, JASA 1989](#)

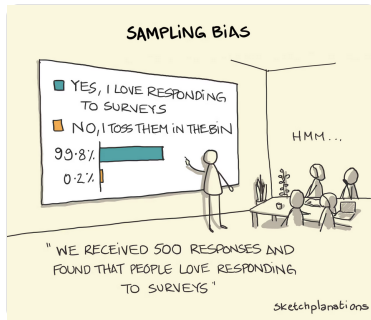


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- **Amazon, 2018.** A recruiting engine trained on ten years of mostly male resumes. It penalized the word “women’s”.

[Dastin, Reuters, 10 Oct 2018](#)

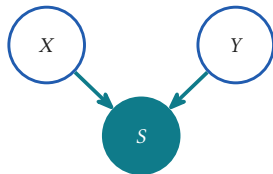


sketchplanations.com/sampling-bias

Selection in the Wild: Colliders

- **Berkson, 1946.** Two unrelated diseases look anti-correlated among hospital patients.

Berkson, Biometrics Bulletin 1946



The same shape, three times.

S: admission, birth weight,
being tested.

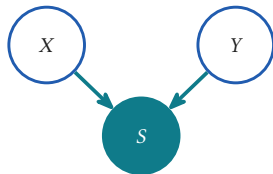
Selection in the Wild: Colliders

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Wilcox, Int. J. Epidemiol. 2001

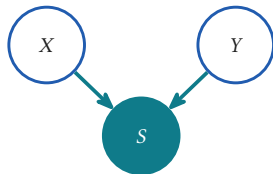


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- **Covid and smoking, 2020.** Smokers looked protected. Who got tested was the collider.
Griffith et al., Nature Communications 2020

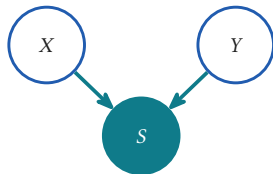


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The same shape, three times.
S: admission, birth weight,
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Always ask what put a row in your table.

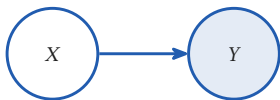
Section 8

Causal and anti-causal scenarios

8

Two scenarios

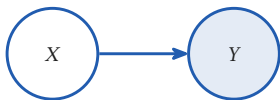
Causal



The features cause the target.

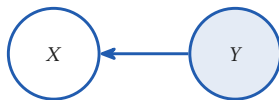
Two scenarios

Causal



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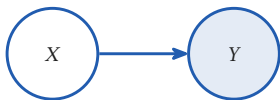
Anti-causal



The target causes the features.

Two scenarios

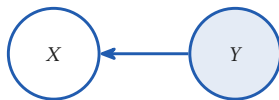
Causal



The features cause the target.

Cancer from location, CO₂,
smoking.

Anti-causal

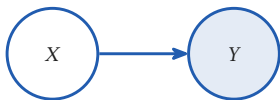


The target causes the features.

Celiac disease from stomachache,
fatigue, income.

Two scenarios

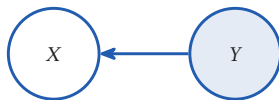
Causal



The features cause the target.

Cancer from location, CO₂,
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Anti-causal



The target causes the features.

Celiac disease from stomachache,
fatigue, income.

Same two nodes. One arrow flipped.

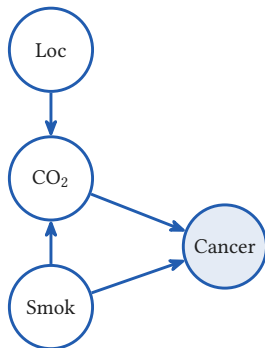
The causal scenario: cancer

- Diagnose cancer.



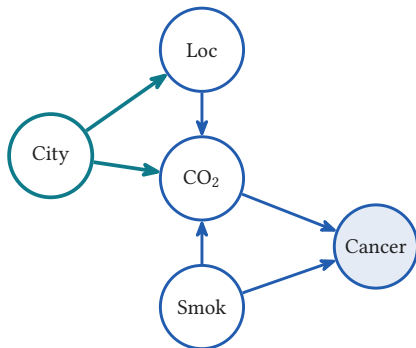
The causal scenario: cancer

- Diagnose cancer.
- Features: location, CO₂ in lungs, smoking.



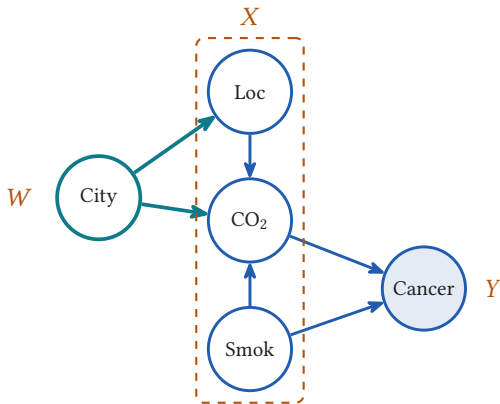
The causal scenario: cancer

- Diagnose cancer.
- Features: location, CO₂ in lungs, smoking.
- Training data: a couple of cities.



The causal scenario: cancer

- Diagnose cancer.
- Features: location, CO₂ in lungs, smoking.
- Training data: a couple of cities.
- The estimator must not depend on the city.



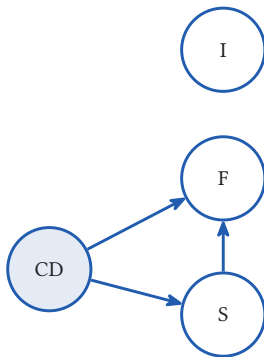
The anti-causal scenario: celiac disease

- Diagnose celiac disease.



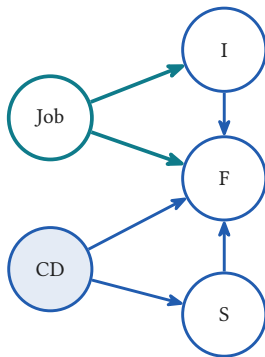
The anti-causal scenario: celiac disease

- Diagnose celiac disease.
- Features: stomachache, fatigue, income.



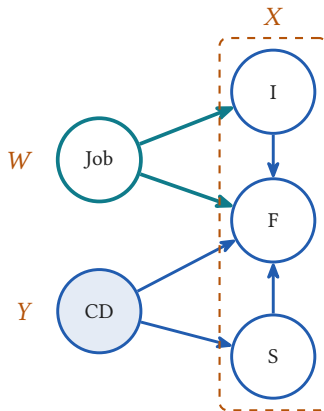
The anti-causal scenario: celiac disease

- Diagnose celiac disease.
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- The estimator must not depend on the job.

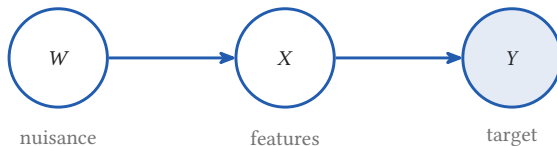


The anti-causal scenario: celiac disease

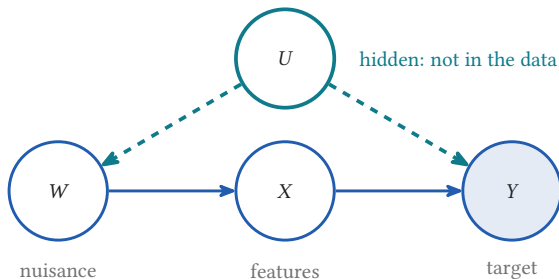
- Diagnose celiac disease.
- Features: stomachache, fatigue, income.
- The estimator must not depend on the job.
- Here the target causes the features.



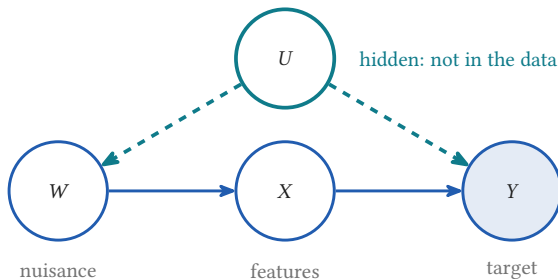
Confounding, formally



Confounding, formally



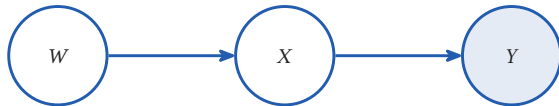
Confounding, formally



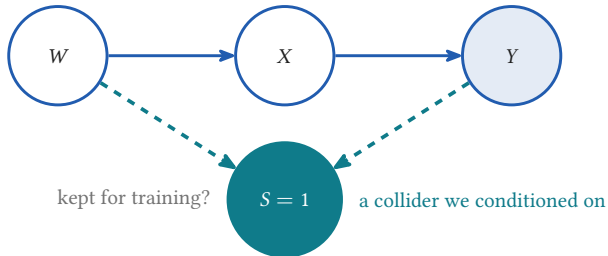
$$W \not\perp Y$$

A hidden common cause creates the correlation.

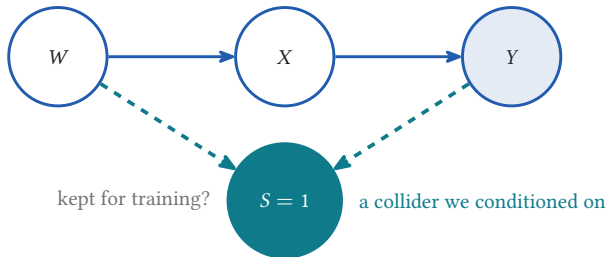
Selection, formally



Selection, formally



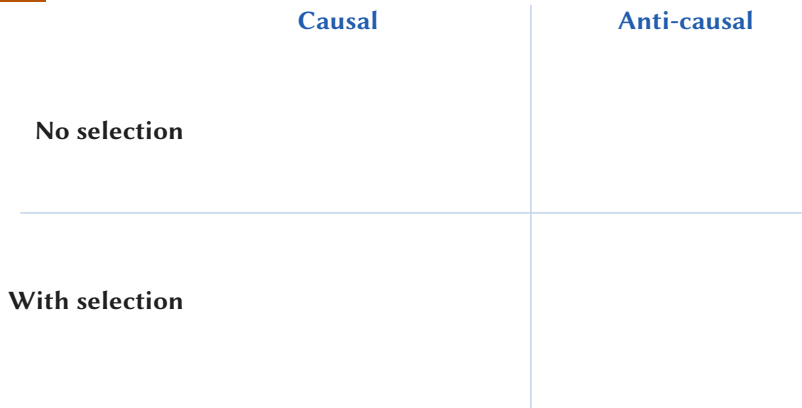
Selection, formally



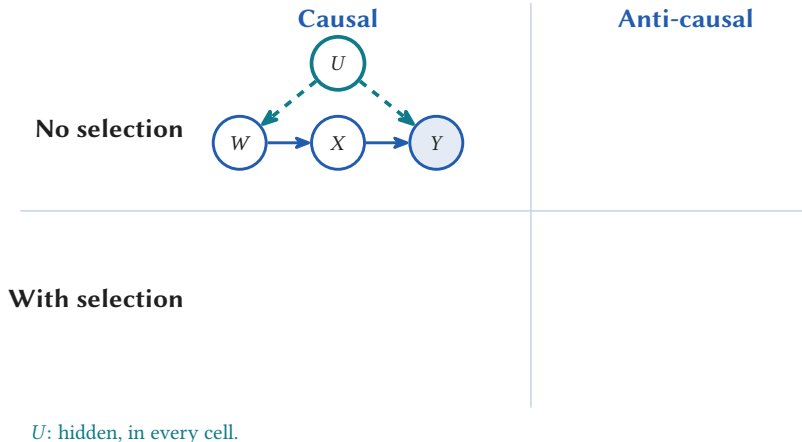
$$W \not\perp Y \mid S = 1$$

Filtering the rows creates the correlation.

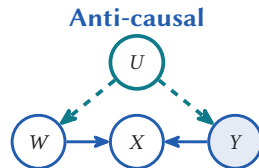
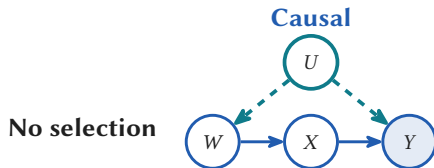
The taxonomy



The taxonomy



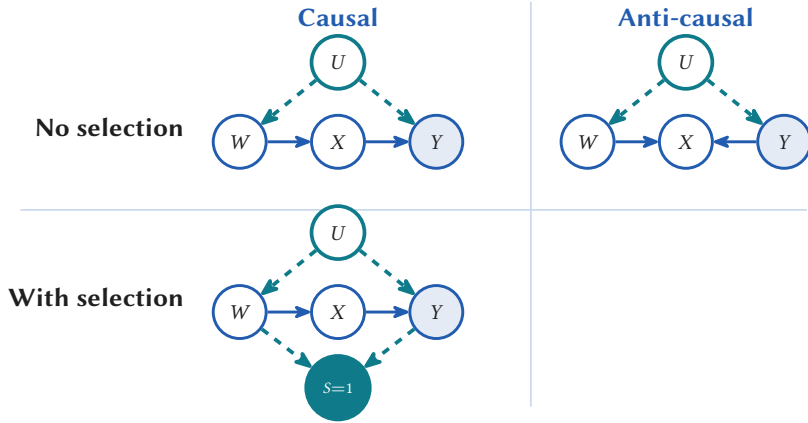
The taxonomy



With selection

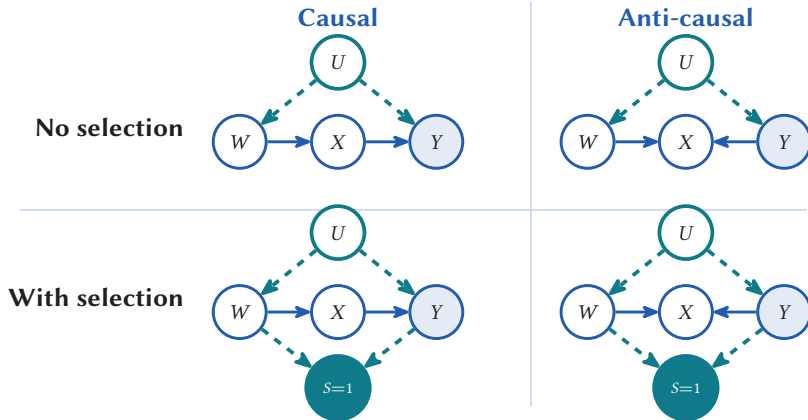
U: hidden, in every cell.

The taxonomy



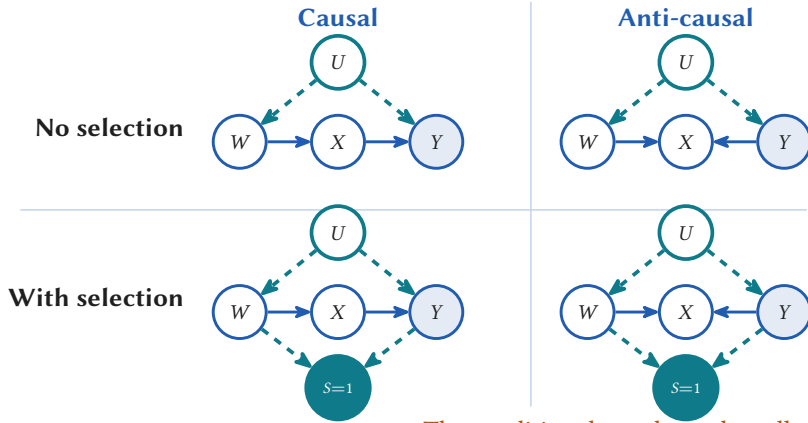
U : hidden, in every cell.

The taxonomy



U : hidden, in every cell.

The taxonomy



U : hidden, in every cell.

The condition depends on the cell.

Section 9

Counterfactual invariance

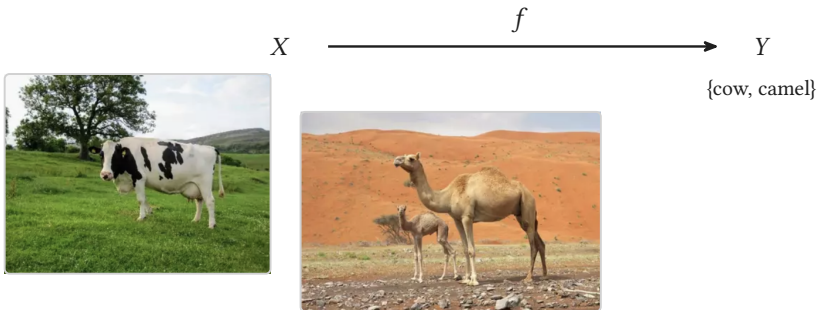
9

Cows and Camels

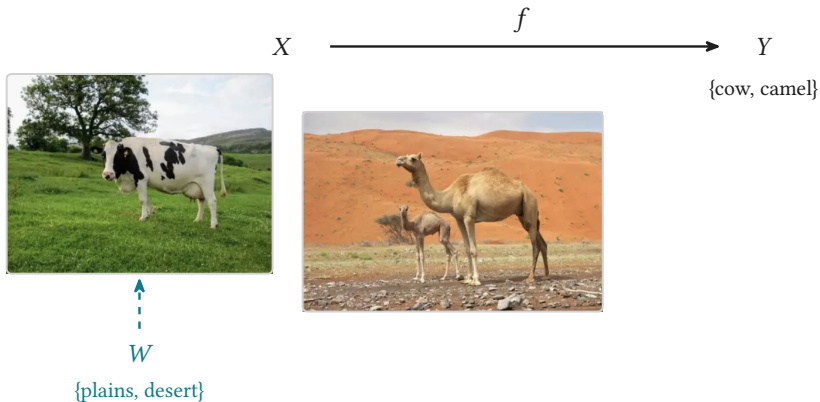
X



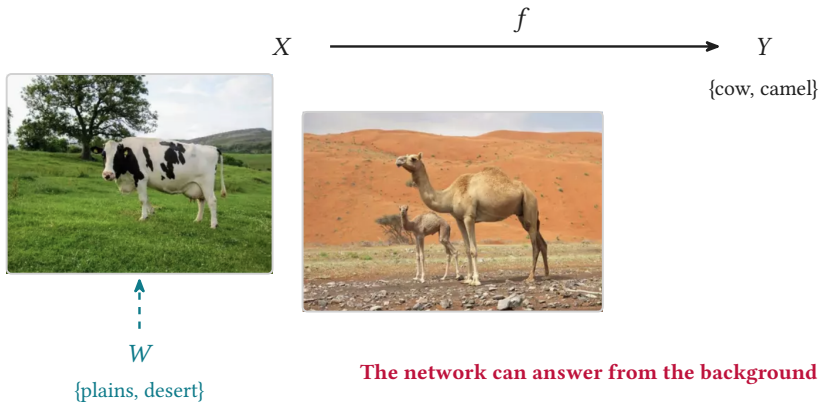
Cows and Camels



Cows and Camels



Cows and Camels



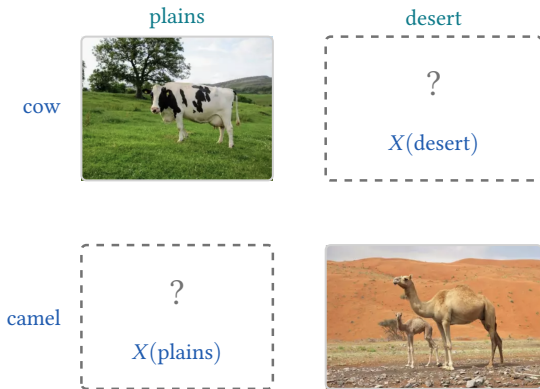
The network can answer from the background alone.

The Counterfactual Question

	plains	desert
cow		
camel		

Rows: the animal (Y). Columns: the environment (W). The training set fills only the diagonal.

The Counterfactual Question



Rows: the animal (Y). Columns: the environment (W). The training set fills only the diagonal.

The Counterfactual Question

	plains	desert	
cow		? $X(\text{desert})$	Force W to be desert. The animal did not change. Does f still say cow?
camel	? $X(\text{plains})$		

Rows: the animal (Y). Columns: the environment (W). The training set fills only the diagonal.

The Counterfactual Question

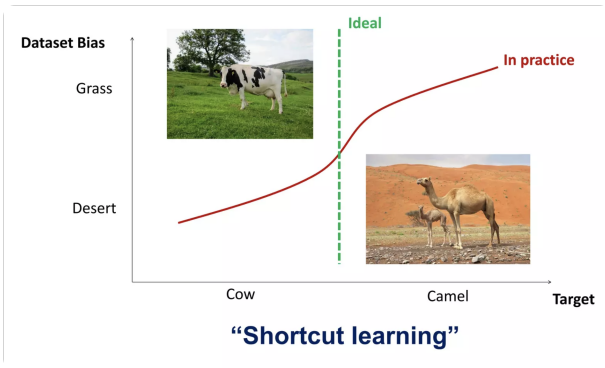
	plains	desert
cow		? $X(\text{desert})$
camel	? $X(\text{plains})$	

Force W to be desert.
The animal did not change.
Does f still say cow?

If not, then f was never using the cow.

Rows: the animal (Y). Columns: the environment (W). The training set fills only the diagonal.

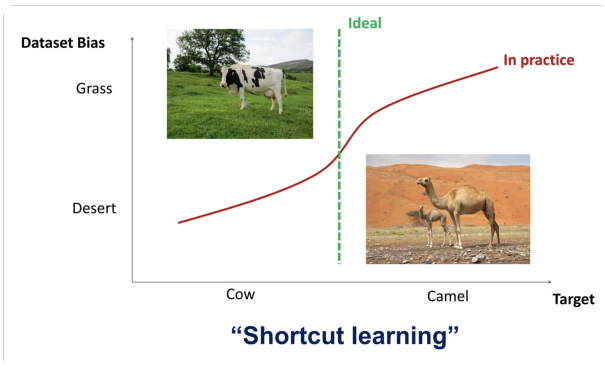
Shortcut Learning



- In training, every cow stands on grass and every camel on sand.

Figure: Park et al., CVPR 2023.

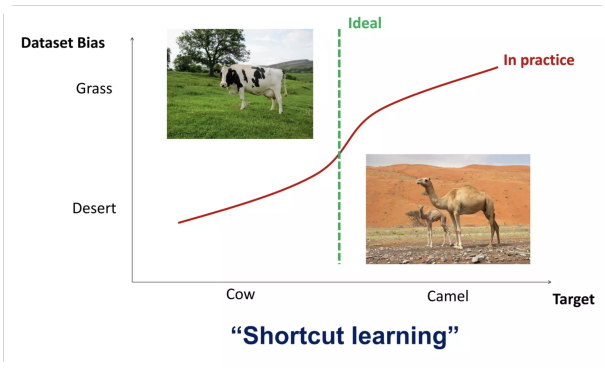
Shortcut Learning



- In training, every cow stands on grass and every camel on sand.
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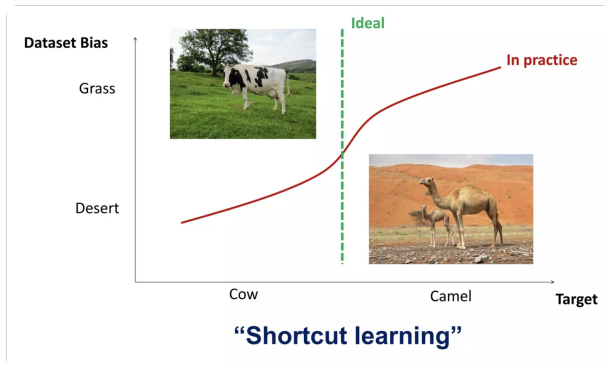
Shortcut Learning



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Figure: Park et al., CVPR 2023.

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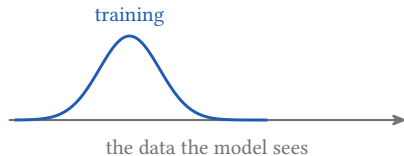
- In training, every cow stands on grass and every camel on sand.
- A spurious correlation between causal and non-causal features.
- The estimator (ab)uses the non-causal feature to generalize.

Right answer, wrong feature.

Figure: Park et al., CVPR 2023.

Domain Shift

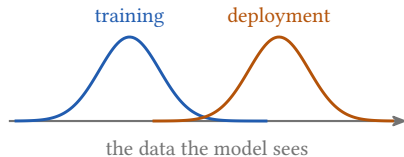
- The test samples come from a different distribution than the training samples.



Wang et al., J. Am. Coll. Radiol. 17(6), 2020. On COVID scans: Roberts et al., Nature Mach. Intell. 3, 2021.

Domain Shift

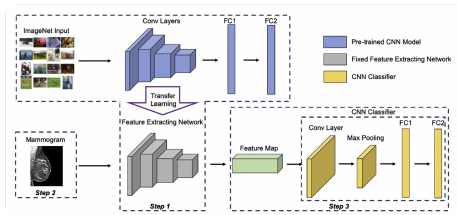
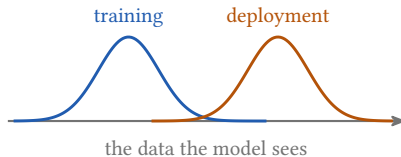
- The test samples come from a different distribution than the training samples.
- Domain shifts happen constantly in practice.



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Domain Shift

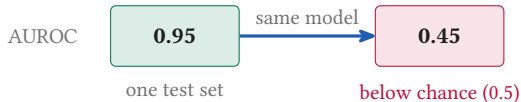
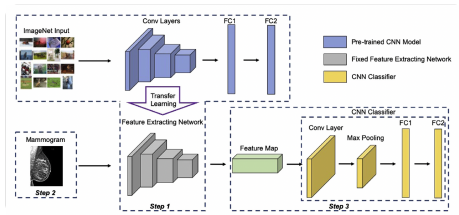
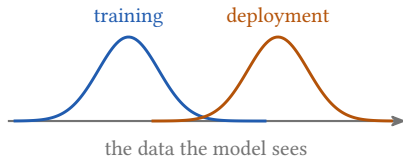
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Domain Shift

- The test samples come from a different distribution than the training samples.
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Wang et al., J. Am. Coll. Radiol. 17(6), 2020. On COVID scans: Roberts et al., Nature Mach. Intell. 3, 2021.

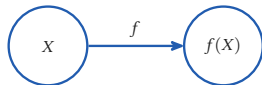
Counterfactual Invariance

- X : the features of an object. Y : the target.



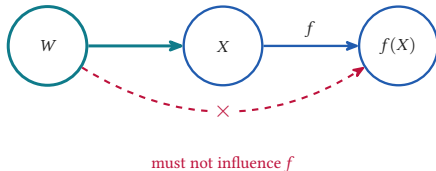
Counterfactual Invariance

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- f : the estimator. It maps X to Y .



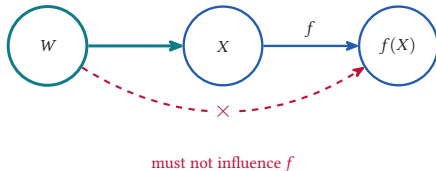
Counterfactual Invariance

- X : the features of an object. Y : the target.
- f : the estimator. It maps X to Y .
- W : it influences X , it must not influence f .
the environment, the scanner, the background, the city, the job



Counterfactual Invariance

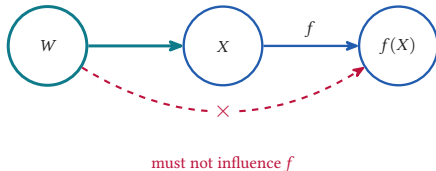
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- $X(w)$: the features we would see, had we set $W = w$.
a counterfactual: we never observed it



Counterfactual Invariance

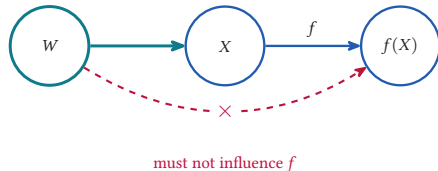
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$$f(X(w))$$



Counterfactual Invariance

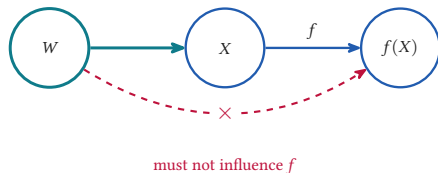
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$$f(X(w)) = f(X(w'))$$

Counterfactual Invariance

- X : the features of an object. Y : the target.
- f : the estimator. It maps X to Y .
- W : it influences X , it must not influence f .
the environment, the scanner, the background, the city, the job
- $X(w)$: the features we would see, had we set $W = w$.
a counterfactual: we never observed it



$$f(X(w)) = f(X(w'))$$

for all $w, w' \in \text{range}(W)$

Invariant Representations



Encoder



Encoder

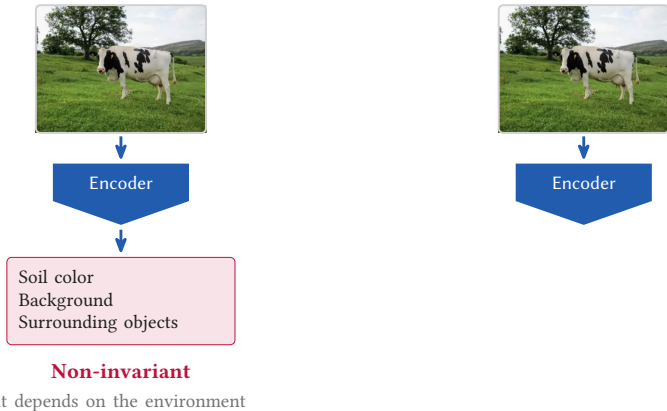
Same picture, two encoders. Only the invariant one survives a new environment.

Invariant Representations



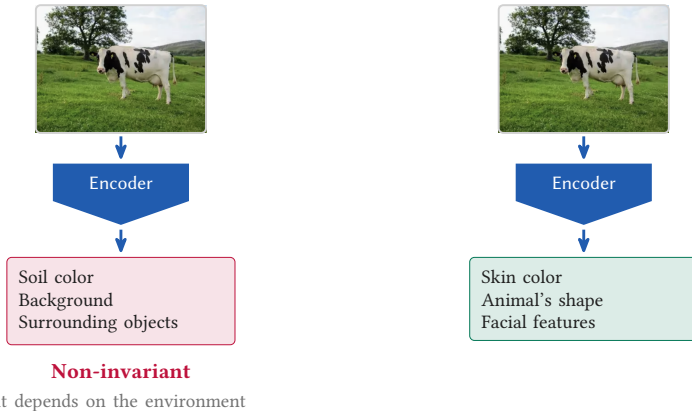
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Invariant Representations



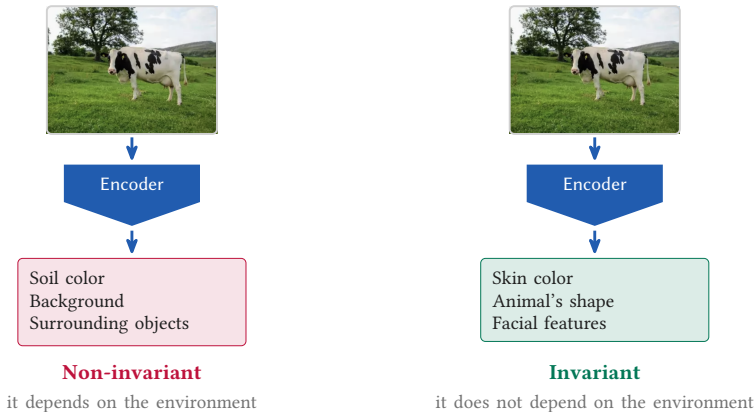
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Invariant Representations



Same picture, two encoders. Only the invariant one survives a new environment.

Invariant Representations



Same picture, two encoders. Only the invariant one survives a new environment.

**We do not just want
a model that is accurate.**

**We want a model that would
give the same answer
in a world where only
the environment changed.**

Section 10

Adversarial training

10

Annotating the environment

x_i

the X-ray



Annotating the environment

x_i the X-ray	y_i the label
	Fracture
	No fracture
	Fracture

Annotating the environment

x_i the X-ray	y_i the label	w_i the environment
	Fracture	Scanner X0862
	No fracture	Scanner Definium 5000
	Fracture	Scanner X5000

Scanner models from the hip fracture study: Badgeley et al., npj Digital Medicine, 2019.

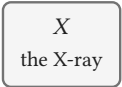
Annotating the environment

x_i the X-ray	y_i the label	w_i the environment
	Fracture	Scanner X0862
	No fracture	Scanner Definium 5000
	Fracture	Scanner X5000

$$\mathcal{D} = \{ (x_i, y_i, w_i) \}_{i=1}^n$$

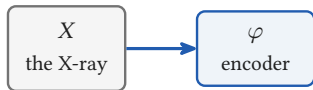
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The architecture

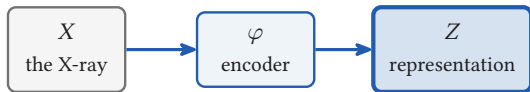


X
the X-ray

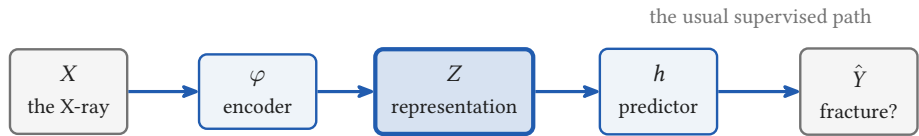
The architecture



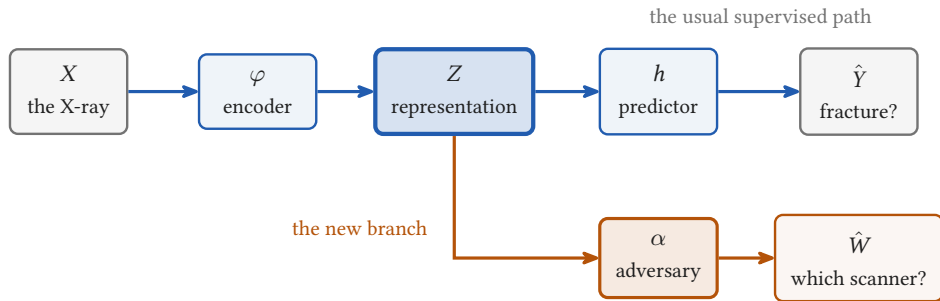
The architecture



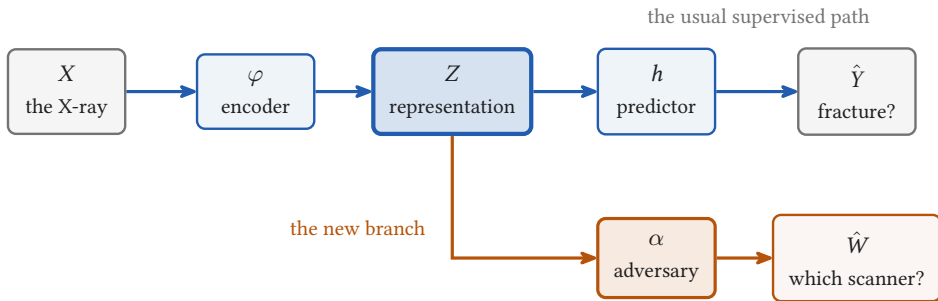
The architecture



The architecture



The architecture



The adversary never sees X . Only Z .

The training objective



Same five steps. We change the loss and the training algorithm. Nothing else.

The training objective



$$\min_{\varphi, h} \mathcal{L}_Y(h(\varphi(X)), Y)$$

The usual risk: the estimator must fit Y .

The training objective



$$\min_{\varphi, h} \mathcal{L}_Y(h(\varphi(X)), Y) - \lambda \mathcal{L}_W(\alpha(\varphi(X)), W)$$

The minus sign: we reward the encoder for making the adversary's job impossible.

The training objective



$$\min_{\varphi, h} \max_{\alpha} \mathcal{L}_Y(h(\varphi(X)), Y) - \lambda \mathcal{L}_W(\alpha(\varphi(X)), W)$$

The adversary plays its best move. The encoder must beat it anyway.

The training objective



$$\min_{\varphi, h} \max_{\alpha} \underbrace{\mathcal{L}_Y(h(\varphi(X)), Y)}_{\text{the estimator must succeed}} - \lambda \underbrace{\mathcal{L}_W(\alpha(\varphi(X)), W)}_{\text{the adversary must fail}}$$

One number, two demands.

The training objective



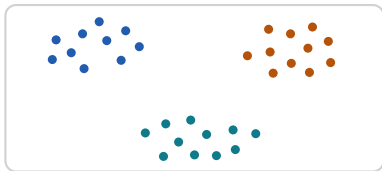
$$\min_{\varphi, h} \max_{\alpha} \underbrace{\mathcal{L}_Y(h(\varphi(X)), Y)}_{\text{the estimator must succeed}} - \lambda \underbrace{\mathcal{L}_W(\alpha(\varphi(X)), W)}_{\text{the adversary must fail}}$$

And λ is the knob.



What this buys us

Without the adversary

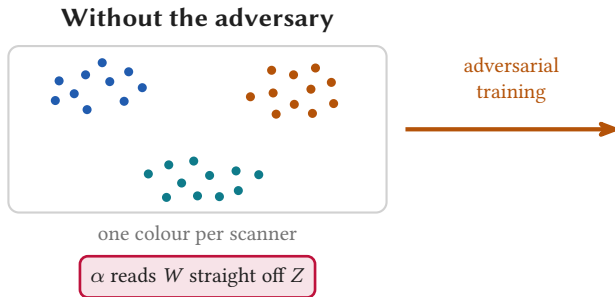


one colour per scanner

α reads W straight off Z

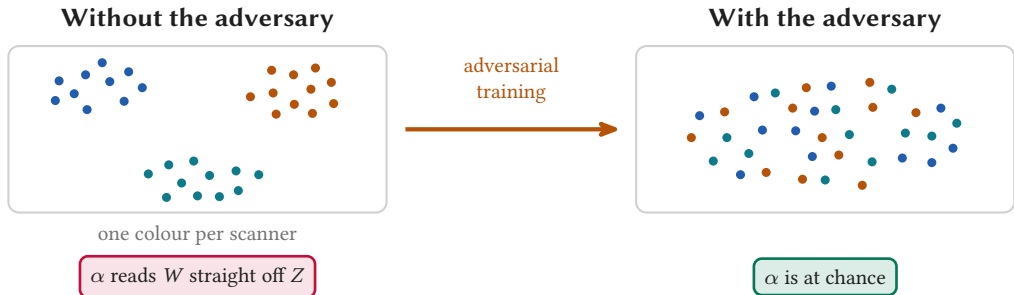
The hip fracture network's features did cluster by scanner model: Badgeley et al., npj Digital Medicine, 2019.

What this buys us



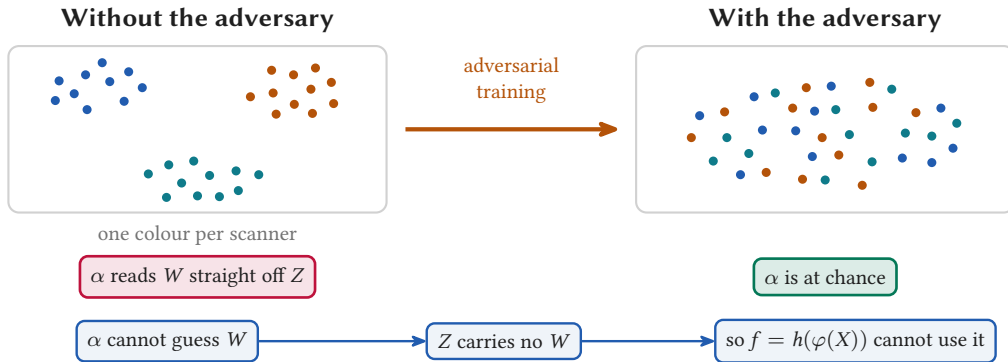
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What this buys us



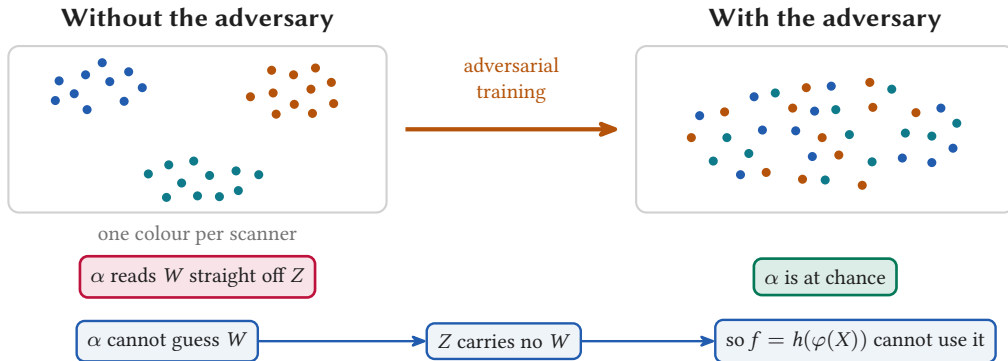
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What this buys us



The hip fracture network's features did cluster by scanner model: Badgeley et al., npj Digital Medicine, 2019.

What this buys us



A proxy, not a proof: a stronger adversary might still find W in Z .

The hip fracture network's features did cluster by scanner model: Badgeley et al., npj Digital Medicine, 2019.

**Counterfactual invariance
is not a property we hope for.
It is a term in the loss.**

Section 11

Necessary conditions for counterfactual invariance



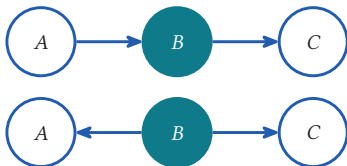
d-separation

Mark a set Z of nodes as observed (filled). The three shapes now decide which paths leak.

d-separation

Mark a set Z of nodes as observed (filled). The three shapes now decide which paths leak.

Non-collision



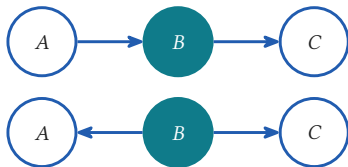
Chain or fork.

Blocked when the middle node **is** observed.

d-separation

Mark a set Z of nodes as observed (filled). The three shapes now decide which paths leak.

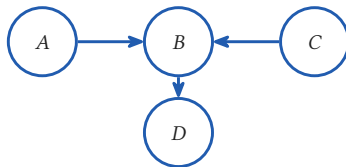
Non-collision



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Collision



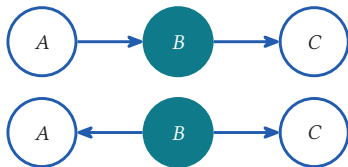
Collider B , with descendant D .

Blocked when both are **not** observed.

d-separation

Mark a set Z of nodes as observed (filled). The three shapes now decide which paths leak.

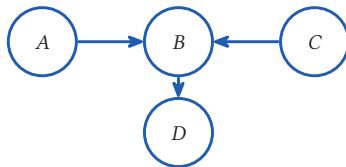
Non-collision



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Collision



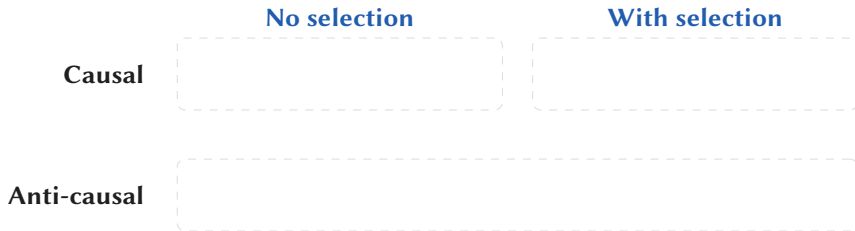
Collider B , with descendant D .

Blocked when both are **not** observed.

Two variables are **conditionally independent**
given Z when every path between them is blocked.

Three Necessary Conditions

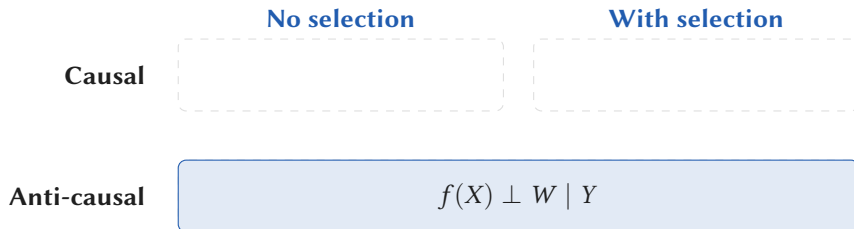
For an estimator f to be counterfactually invariant, the following is needed.



Veitch et al., “Counterfactual Invariance to Spurious Correlations”, NeurIPS 2021. We sketch the first two cases; the third is in the paper.

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For an estimator f to be counterfactually invariant, the following is needed.

	No selection	With selection
Causal	$f(X) \perp W$	
Anti-causal	$f(X) \perp W \mid Y$	

Veitch et al., “Counterfactual Invariance to Spurious Correlations”, NeurIPS 2021. We sketch the first two cases; the third is in the paper.

Three Necessary Conditions

For an estimator f to be counterfactually invariant, the following is needed.

	No selection	With selection
Causal	$f(X) \perp W$	$f(X) \perp W \mid Y$ under one extra assumption
Anti-causal	$f(X) \perp W \mid Y$	

Veitch et al., “Counterfactual Invariance to Spurious Correlations”, NeurIPS 2021. We sketch the first two cases; the third is in the paper.

Three Necessary Conditions

For an estimator f to be counterfactually invariant, the following is needed.

	No selection	With selection
Causal	$f(X) \perp W$	$f(X) \perp W \mid Y$ under one extra assumption
Anti-causal	$f(X) \perp W \mid Y$	

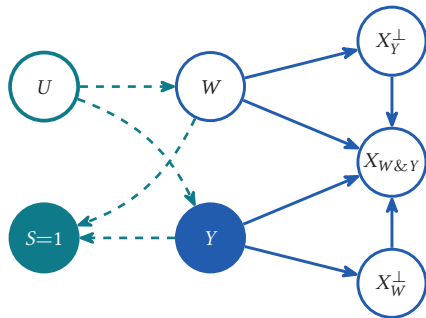
The taxonomy from before, one condition per cell.

Veitch et al., “Counterfactual Invariance to Spurious Correlations”, NeurIPS 2021. We sketch the first two cases; the third is in the paper.

Proof Sketch

Anti-causal scenario, with selection.

- An invariant f can use only $X_W^\perp$.

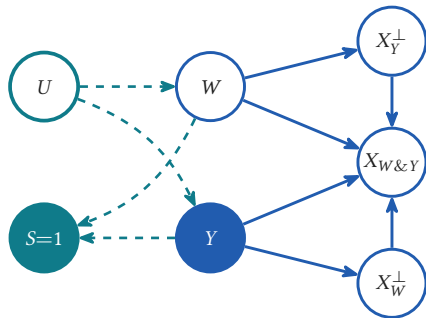


Filled: observed. We only see rows with $S = 1$.
 U is hidden.

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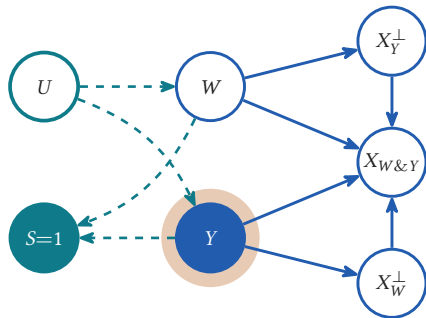


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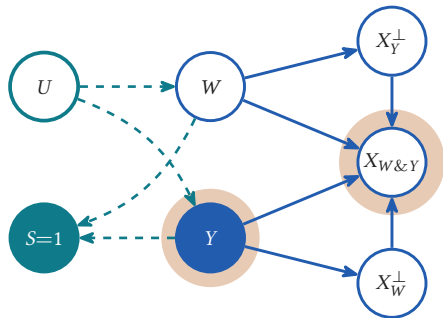


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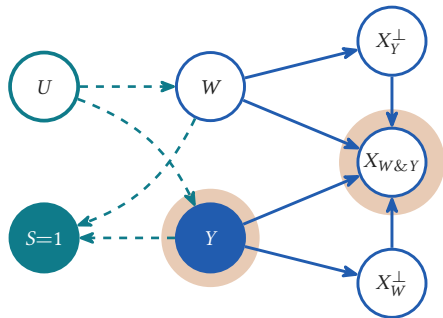
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Every path is blocked. Read it off the graph.



Filled: observed. We only see rows with $S = 1$.
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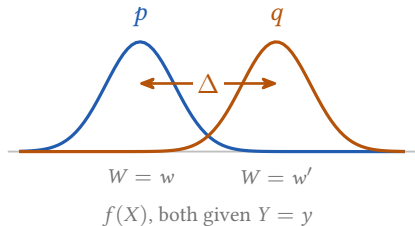
Enforcing Counterfactual Invariance

Restrict, without loss of generality, to the anti-causal scenario, with W and Y binary. We must ensure $f(X) \perp W \mid Y$: these two distributions must be equal, for any w, w' and any y .

$f(X)$, both given $Y = y$

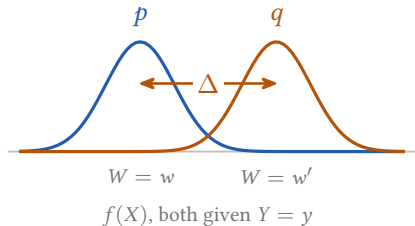
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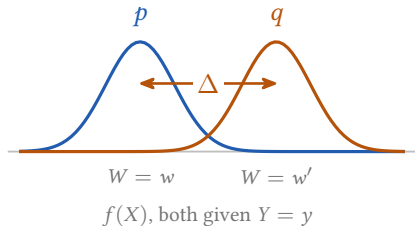
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$$\underbrace{P(f(X) \mid W = w, Y = y)}_p \quad \underbrace{P(f(X) \mid W = w', Y = y)}_q$$

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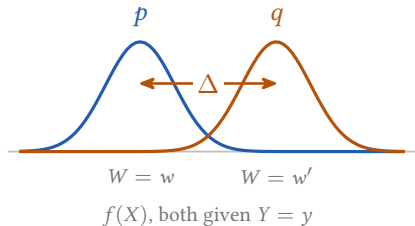
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Choose a distance Δ between distributions.

$$\Delta \left(\underbrace{P(f(X) \mid W = w, Y = y)}_p, \underbrace{P(f(X) \mid W = w', Y = y)}_q \right)$$

Enforcing Counterfactual Invariance

Restrict, without loss of generality, to the anti-causal scenario, with W and Y binary. We must ensure $f(X) \perp W \mid Y$: these two distributions must be equal, for any w, w' and any y .



1

Choose a distance Δ between distributions.

2

Force p to equal q by minimizing $\Delta(p, q)$.

$$\min_f \Delta \left(\underbrace{P(f(X) \mid W = w, Y = y)}_p, \underbrace{P(f(X) \mid W = w', Y = y)}_q \right)$$

Back to Adversarial Training

What to enforce

$p: f(X) \text{ given } W = w$



$q: f(X) \text{ given } W = w'$

Both given $Y = y$. Make them coincide.

How to enforce it

Back to Adversarial Training

What to enforce

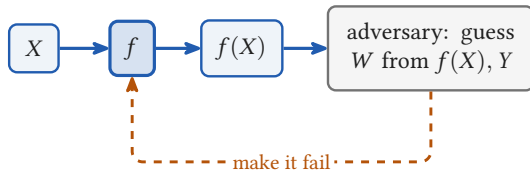
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The adversary measures how far p is from q .

Back to Adversarial Training

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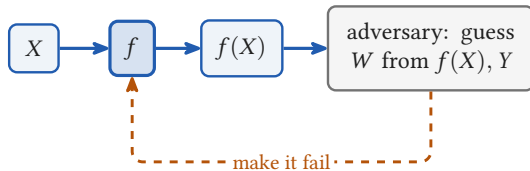
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Beating the adversary is minimizing Δ .

Back to Adversarial Training

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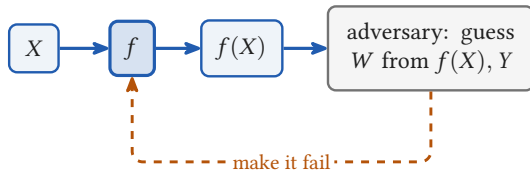
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How to enforce it



The adversary measures how far p is from q .

Beating the adversary is minimizing Δ .

The theorem says what. Adversarial training says how.

Summary



The ice is real

hip fractures, mammograms, Mary's contacts

Summary

1

The ice is real

hip fractures, mammograms, Mary's contacts

2

The graph says where it is

chain, fork, collider: confounding, selection, d-separation

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Invariance keeps you off it

$f(X) \perp W \mid Y$, enforced by an adversary

Summary

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The ice is real

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The graph says where it is

chain, fork, collider: confounding, selection, d-separation

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Invariance keeps you off it

$f(X) \perp W \mid Y$, enforced by an adversary

A model can be right for the wrong reason.

This weekend

1

Recap

where Weekend 1 left us

2

Confounding and selection

the two ways data misleads a model

3

Explainable AI: attribution maps

tools to see what a model used

4

Prompt optimization (GEPA)

a separate module